A PTAS FOR TSP WITH NEIGHBOURHOODS OVER PARALLEL LINE SEGMENTS OF SIMILAR LENGTH

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Allowed standard prerequisites (editorial).

- Arora portal-respecting hierarchical TSP dissection and patching. Author bibliography Arora98 and precise invocations in native proof; author-specific discretization, shadow bounds, and DP all retained.
- Planar geometry, Eulerian traversal and dynamic programming. Complete author local lemmas retained, including deferred proofs from conference abstract.

1. Introduction

The Travelling Salesman Problem (TSP) is one of the most fundamental and well-studied problems in combinatorial optimization due to its wide range of applications. In TSP, one is given a set of points in a metric space and the goal is to find a (closed) tour (or walk) of minimum length visiting all the points. For several decades, the classic algorithm by Christofides Christofides (1976) and independently by Serdyukov Serdyukov (1978) which implies a $\frac{3}{2}$-approximation was the best-known approximation for TSP until a recent result in Karlin et al. (2022) which shows a slight improvement. Several generalizations (or special cases) of TSP have been studied as well, the most notable is when the points are given in fixed dimension Euclidean space. Arora and Mitchell Arora (1998); Mitchell (1999) presented different PTASs for (fixed dimension) Euclidean TSP. There have been many papers that have extended these results. Arkin and Hassin Arkin and Hassin (1994) introduced the notion of TSP with neighbourhoods (TSPN).

An instance of TSPN is a set of neighbourhoods (or regions) given in a metric space, and the goal is to find a minimum length (or cost) tour that visits all these regions. Each region can be a single point or could be defined by a subset of points. They gave several $O(1)$-approximations for the geometric settings where each region is some well-defined shape on the plane, e.g. disks, and parallel unit length segments. Several papers have studied TSPN for various classes of neighbourhoods and under different metrics.

TSPN is much more difficult than TSP in general and in special cases, just as group Steiner tree is much more difficult than Steiner tree (one can consider each neighbourhood as a group/set from which at least one point needs to be visited). In group Steiner tree or group TSP, one is given a metric along with groups of terminals (each group is a finite set). The goal is to find a minimum cost Steiner tree (or a tour) that contains (or visits) at least one terminal from each group. TSPN generalizes group TSP by allowing infinite size groups. Using the hardness result for group Steiner tree Halperin and Krauthgamer (2003), it follows that general TSPN is hard to approximate within a factor better than $\Omega(\log^{2-\varepsilon} n)$ for any $\varepsilon > 0$ even on tree metrics. The algorithms for group Steiner tree on trees in Garg et al. (1998), and embedding of metrics onto tree metrics in Fakcharoenphol et al. (2004), imply an $O(\log^3 n)$-approximation for TSPN in general metrics. Unlike Euclidean TSP (which has a PTAS), TSPN is **APX**-hard on the Euclidean plane (i.e. $\mathbb{R}^2$) de Berg et al. (2005). The special case when each region is an arbitrary finite set of points in the Euclidean plane (group TSP) has no constant approximation Safra and Schwartz (2003) and the problem remains **APX**-hard even when each region consists of exactly two points Dror and Orlin (2008).

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Focusing on Euclidean metrics, most of the earlier work have studied the cases where the regions (or objects) are *fat*. Roughly speaking, it usually means the ratio of the smallest enclosing circle to the largest circle fitting inside the object is bounded. There are some work on when regions are *not* fat, most notably when the regions are (infinite) lines or line segments or in higher dimensions when they are hyperplanes. For the case of infinite line segments on $\mathbb{R}^2$, the problem for n lines can be solved exactly in $O(n^4 \log n)$ time by a reduction to the Shortest Watchman Route Problem (see Dror et al. (2003); Jonsson (2002)). For the same setting, Dumitrescu and Mitchell Dumitrescu and Mitchell (2003) presented a linear time $\frac{\pi}{2}$-approximation, which was improved to $\sqrt{2}$ by Jonsson Jonsson (2002) (again in linear time). For infinite lines in higher dimensions (i.e. ≥ 3), the problem is proved to be **APX**-hard (see Antoniadis et al. (2022) and references there). For neighbourhoods being hyperplanes and dimension being $d \geq 3$, Dumitrescu and Tóth Dumitrescu and Tóth (2016) present a constant factor approximation (which grows exponentially with d). For arbitrary d, they present an $O(\log^3 n)$-approximation. For any fixed $d \geq 3$, authors of Antoniadis et al. (2020) present a PTAS.

For parallel (unit) line segments on the plane ($\mathbb{R}^2$) Arkin and Hassin Arkin and Hassin (1994) presented a $(3\sqrt{2}+1)$-approximation which was improved to $3\sqrt{2}$ by Dumitrescu and Mitchell (2003), and this remains the best-known approximation for this case as far as we know for over two decades. Authors of Elbassioni et al. (2009) proved that TSPN for unit line segments (in arbitrary orientation) is **APX**-hard. In this paper, we settle the approximability of TSPN when regions are parallel line segments of similar length (unit length is a special case) and present a PTAS for it. We first focus on the case of unit line segments and show how our result extends to when line segments have bounded length ratio. This is in contrast with the **APX**-hardness of Elbassioni et al. (2009) for unit line segments with arbitrary orientation. Our result also implies a $(2+\varepsilon)$-approximation for the case where the segments are axis-parallel (i.e. can be both vertical and horizontal).

1.1. **Related Work.** The work on TSPN is extensive, we list a subset of the most notable and relevant work here and refer to the references of them for earlier works. All works listed are for $\mathbb{R}^2$ metric. Arkin and Hassin Arkin and Hassin (1994) presented constant factor approximations for several TSPN cases including when the neighbourhoods are parallel unit-length line segments (with ratio $3\sqrt{2}+1$). Very recently, PTASs were proposed for the case of unit disks and unit squares in Bandyapadhyay et al..

Mata and Mitchell Mata and Mitchell (1995) presented $O(\log n)$-approximation for general connected polygonal neighbourhoods. Mitchell Mitchell (2007) gave a PTAS the case that regions are *disjoint* fat objects. This built upon his earlier work on PTAS for Euclidean TSP Mitchell (1999). However, it is still open whether the problem is **APX**-hard for disjoint (general) shapes.

Dumitrescu and Mitchell Dumitrescu and Mitchell (2003) presented several results, including $O(1)$-approximation for TSPN where objects are connected regions of the same or similar diameter. This implies $O(1)$-approximation for line segments of the same length in arbitrary orientation. They also improved the bounds for cases of parallel segments of equal length, translates of a convex region, and translates of a connected region. For parallel (unit) line segments they obtain a $3\sqrt{2}$-approximation which remained the best ratio for this class until now.

Feremans and Grigoriev Feremans and Grigoriev (2005), gave a PTAS for the case that regions are disjoint fat polygons of similar size. A similar result was obtained by Bodlaender et al. (2009) which gave a PTAS for TSPN for disjoint fat regions of similar sizes. Mitchell Mitchell (2010) presented an $O(1)$-approximation for planar TSPN with pairwise-disjoint connected neighbourhoods of any size or shape (see also de Berg et al. (2005)). Subsequently, Elbassioni et al. (2005) gave $O(1)$-approximation for discrete setting where regions are overlapping, convex, and fat with similar size.

These results were further generalized by Chan and Elbassioni Chan and Elbassioni (2011) who presented a QPTAS for fat weakly disjoint objects in doubling dimensions (this allows limited amount of overlapping of the objects). This was further improved to a PTAS for the same setting Chan et al. (2016). The TSPN problem is even harder if the neighbourhoods are disconnected. See the surveys of Mitchell Mitchell (2000, 2004). If the metric is planar and each group (or neighbourhood) is the set of vertices around a face, Bateni et al. (2016) presents a PTAS for the group Steiner tree and a $(2+\varepsilon)$-approximation for TSPN.

1.2. **Our Results and Technique.** The main result of this paper is the following theorem.

Theorem 1. *Given a set of n parallel (say vertical) line segments with lengths in $[1, \lambda]$ for a fixed λ as an instance of TSPN, there is an algorithm that finds a $(1+\varepsilon)$-approximation solution in time $n^{O(\lambda/\varepsilon^3)}$.*

The algorithm we present is randomized but can be easily derandomized (like the PTAS for classic TSP). To simplify the presentation, we give the proof for the case of unit line segments first.

This problem generalizes the classic (point) TSP (at a loss of $(1+\varepsilon)$ factor). Given a point TSP instance, scale the plane so that the minimum distance between the points is at least $1/\varepsilon$; call this instance $\mathcal{I}$. Obtain instance $\mathcal{I}'$ for line TSP by placing a vertical unit line segment over each point. It is easy to see that the optimum solutions of $\mathcal{I}$ and $\mathcal{I}'$ differ by at most an ϵ factor.

The difficult cases for line TSP are when the line segments are not too far apart (for e.g. they can be packed in a box of size $O(\sqrt{n})$ or smaller). There are two key ingredients to our proof that we explain here. One may try to adapt the hierarchical decomposition by Arora Arora (1998) to this setting. Following that hierarchical decomposition, the first issue is that some line segments might be crossing the horizontal dissecting lines, and so we don't have independent sub-instances, and it is not immediately clear in which subproblem these crossing segments must be covered by the tour. Note that the number of line segments crossing a dissecting line can be large. Our first insight is the following:

Insight 1. *At a loss of $(1+\varepsilon)$, we can drop the line segments crossing horizontal dissecting lines and instead requiring a subset of portals of each square (of the dissection) to be visited, provided we continue the quad-tree decomposition until each square has size $\Theta(1/\varepsilon)$.*

In other words, assuming all the squares in the decomposition have height at least $\Omega(1/\varepsilon)$, then at a small loss we can show that a solution for the modified instance where line segments on the boundary of the squares are dropped, can be extended to a solution for the original instance. So, proving this property allows us to work with the hierarchical (quad-tree) decomposition until squares of size $\Theta(1/\varepsilon)$. This can be proved by a proper packing argument. But then we need to be able to solve instances where the height of the sub-problem instance is bounded by $O(1/\varepsilon)$.

Let's define the notion of shadow of a solution (or in general, shadow of a collection of paths on the plane) as the maximum number of times a vertical sweeping line Γ (that moves from left to right) intersects any of these paths. Our second insight is the following:

Insight 2. *If we consider a window that is a horizontal strip of height $O(h)$ and move this window vertically anywhere over an optimum solution, then the shadow of the parts of optimum visible in this strip is at most $O(h)$.*

In other words, one expects that in the base case of the decomposition (where squares have height $\Theta(1/\varepsilon)$), the shadow is bounded by $O(1/\varepsilon)$. Despite our efforts, proving this appears to be more difficult than thought and it seems there are examples where, even in the unit length segments, the shadow may be large (see Figure 1). We were not able to prove this nor come up with an explicit counterexample. However, we are able to prove the following slightly weaker version that still allows us to prove the final result:

Insight 2 (Revised)**.** *There is a $(1+\varepsilon)$-approximate solution such that the shadow of any strip of height h over that solution is bounded by $O(f(\varepsilon)\cdot h)$ for some function $f(\cdot)$.*

The proof of this insight forms the bulk of our work. We characterize specific structures that would be responsible for having a large shadow in a solution and show how we can modify the solution to a $(1+\varepsilon)$-approximate one with shadow $O(1/\varepsilon)$. Consider the unit line segments case, and suppose opt is the cost of an optimum solution.

Theorem 2. *Given any $\varepsilon > 0$, there is a solution $\mathcal{O}'$ of cost at most $(1+\varepsilon)\cdot$ opt such that in any strip of height 1, the shadow of $\mathcal{O}'$ is $O(1/\varepsilon)$.*

Proof of this theorem appears in Subsection 2.3. We will show that this near-optimum solution has in fact further structural properties that allow us to solve the bounded height cases at the base cases of the hierarchical decomposition using a Dynamic Program (DP) which, later on, is referred to as the inner DP. Proof of Theorem 2 is fairly long and involves multiple steps that gradually proves structural properties for specific configurations.

To get a very high level idea of the proof of this theorem, consider some fixed optimum solution OPT. We decompose the problem into horizontal sections by drawing horizontal lines, called *cover-lines* that are 1-unit apart. The region of the plane between two consecutive cover-lines is called a *strip*. Note that each line segment crosses one cover-line (or might touch exactly two consecutive cover-lines). Let's consider one strip, say S, and consider the intersection of OPT with this strip. This intersection looks like a collection of paths that enter/exit this strip. We define the shadow of this strip similarly: consider a hypothetical vertical

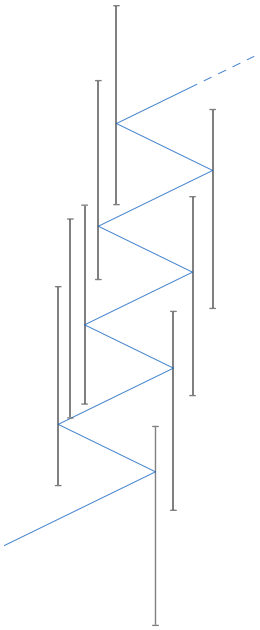


FIGURE 1. A potential arrangement of line segments where the solution has a large shadow

sweep line that moves left to right along the x-axis, the maximum number of intersections of this sweep line with these pieces of OPT restricted to S is the shadow in S. We show that we can modify OPT to a near-optimum solution of cost at most $(1+\varepsilon)$ times that of OPT so that the shadow in each strip is at most $O(1/\varepsilon)$. To prove this, we show that there are certain potential structures that can cause OPT having a large shadow in a strip, one of which we call a zig-zag (see Figure 8). We show that we can modify OPT (at a small increase to the cost) so that the shadow becomes bounded along each zig-zag (or similar structures).

1.2.1. *Case Study: Restricted Special Instances.* To get an intuition of our approach, consider a special case where the given instance of the problem fits within a bounding box of height at most 3. If we draw a horizontal cover-line going through the bottom-most point of the top-most segment, and draw a second cover-line 1 unit below the previous one, then one can see that the entire optimum solution should fit within this strip of height 1 between these two cover-lines. Every segment of the instance intersects with at least one of the two horizontal lines. We call the segments intersecting the top horizontal line *top segments*, and the rest *bottom segments*. Any optimum solution can be partitioned into two subpaths: One subpath from the left-most point on the optimum solution towards the right-most point on the solution (call P_1), and one subpath the opposite way (call P_2), where P_1 is always "above" P_2 and is responsible of covering the top segments, and P_2 is responsible for covering the bottom segments. In this case the shadow of the solution is always 2. This is similar to the classic bitonic TSP tour which can be easily solved using a dynamic program (where there is a sweeping line moving left to right and keeps track of the end-points of these paths P_1 and P_2).

When the height of the bounding box is larger than 3, we still draw horizontal cover-lines that are 1-unit apart. In Figure 3 you can see an instance with a bounding box of height 5 and a solution for it.

If we can prove that the overall shadow of an optimum solution is bounded in every strip, then we can create an extension of the bitonic tour solution to obtain a dynamic program with low complexity (i.e. constant number of subpaths), to then find an exact solution by combining the subpaths we find.

Unfortunately, it might be the case that the shadow of an optimum solution is much larger than a function of the height of the bounding box (i.e. the number of strips). Such configurations like Figure 1 might happen, leading to large shadows. We will show that such configurations can be changed to another solution with a bounded shadow at a small increase in cost.

Organization of the paper: We start by some preliminaries in the next subsection. In Section 2, we first list the properties of an optimum solution (in Subsection 2.1), then the properties of a structured near-optimum

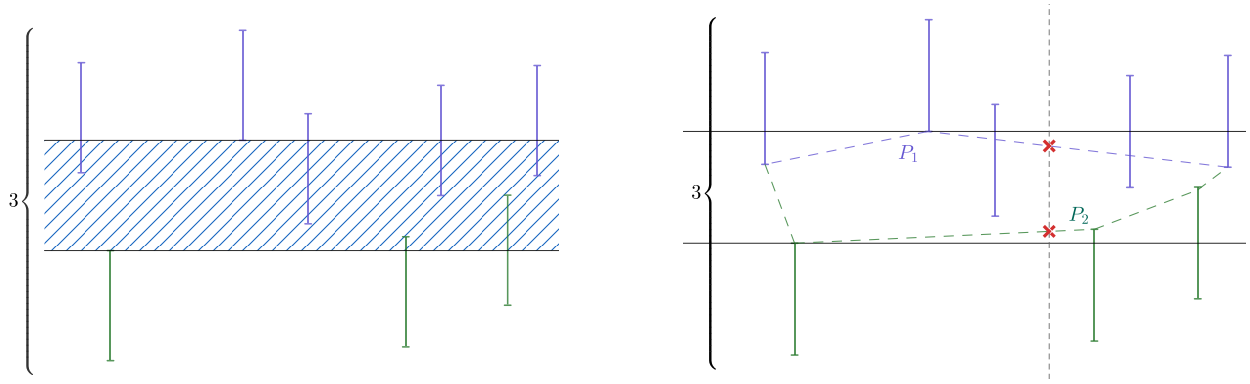


FIGURE 2. An optimum solution to these special case instances intersects with any vertical line at most twice

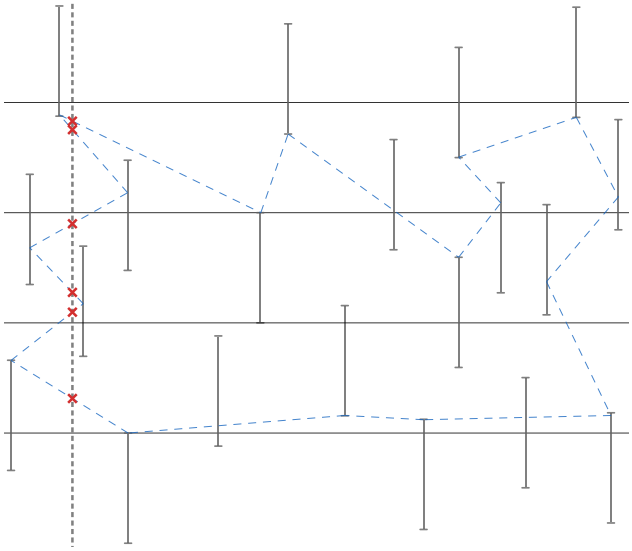


FIGURE 3. An example of an instance of the problem with bounding box of height at most 5. We have at most 3 strips in this case and maximum shadow of the given solution is 6

solution (in Subsection 2.2). In these two sections we list four major lemmas (Lemmas 8 to 11) that are used to prove Theorem 2. We then prove Theorem 2 in Subsection 2.3.

We describe the main algorithm in Section 3, which includes the outer DP and inner DP. The proof of our main theorem appears in Subsection 3.3.

Section 4 has details of the proofs of the major lemmas (for properties of a near-optimum solution), as well as the proof of Theorem 3. Finally, we conclude our results in Section 5 and discuss further extensions.

1.3. **Preliminaries.** Suppose we are given n vertical line segments $s_1, \ldots, s_n$ of lengths in the range $[1, \lambda]$ for some constant $\lambda \geq 1$, and the top and bottom points of each s_i are denoted by s_i^t and s_i^b, respectively. These end-points are also called *tips* of the segment. For any point p, let $x(p)$ and $y(p)$ denote the x and y-coordinates of p, respectively. Similarly, for any segment or vertical line s, let $x(s)$ denote its x-coordinate. For two points p, q, we use $||pq||$ to denote the Euclidean distance between them. A feasible (TSP) tour is specified by a sequence of points where each of these points is on one of the segments of the instance and each line segment has at least one such point, and the tour visits these points consecutively using straight lines. The line that connects two consecutive points in a tour is called a *leg* of the tour. In our problem, the goal is to find a TSP tour of minimum total length. As mentioned earlier, we focus on the case where all the line segments have length 1 and then show how the proof easily extends to the setting where they have lengths in $[1, \lambda]$. Fix an optimum solution, which we refer to by OPT and use opt to refer to its cost. Our

goal is to show the existence of a near-optimum (i.e. $(1+\varepsilon)$-approximate) structured solution that allows us to find it using dynamic programming.

First, we show at a small loss we can assume all the line segments have different x-coordinates. We assume that the minimal bounding box of these line segments has length L and height H. For now, assume $H > 3$ (case of $H \leq 3$ is easier, see Theorem 3). Let $B = \max\{L, H-2\}$. So opt $\geq 2B$; we can also assume $B \leq \frac{n}{\varepsilon}$, because otherwise opt $\geq 2n/\varepsilon$ and if we consider an arbitrary point on each line segment (say the lower tip) and use a PTAS for the classic TSP for these points, then it will be a PTAS for our original instance as well (because we pay at most an extra $+2$ for each line for a total of $2n$ which is $O(\varepsilon \cdot \text{opt})$). For a given $\epsilon > 0$, consider a grid on the plane with side length $\frac{\epsilon B}{n^2}$. Now move each line segment (parallel to the y-axis) so that the lower tip of each s_i is moved to the nearest grid point where there is no other line segment s_i with that x-coordinate. By doing this, all the segments will have different x-coordinates and each segment would move at most $\frac{\sqrt{2}}{2} \cdot \frac{\varepsilon B}{n} < \frac{\epsilon B}{n}$, and in total, all segments would move at most a distance of ϵB. So the optimum value of the new instance has cost at most $(1+\varepsilon) \cdot \text{opt}$. For simplicity of notations, from now on we assume the original instance has this property and let OPT (and opt) refer to an optimum (and its value) of this modified instance.

2. Properties of a Structured Near-Optimum Solution

We start by presenting some properties of an optimum solution and then some structural properties for a near optimum that allow us to prove Theorem 2.

2.1. Properties of an Optimum Solution.

We start by stating some lemmas that give a better understanding of the geometrical properties of an optimum solution, and later build up the proof one of the major lemmas in Subsection 4.1.1 (Lemma 8) from these properties.

One special instance of the problem is when there is a single horizontal line that crosses all the input segments. This special case can be detected and solved easily. Otherwise, any optimum solution will visit at least 3 points that are not colinear. In such cases, like in the classic (point) TSP Arora (1998), we can assume the optimum does not cross itself, i.e. there are no two legs of the optimum ℓ (between points p, q) and ℓ' (between points p', q') that intersect, as otherwise removing these two and adding the pair of $pq', p'q$ or pp', qq' will be a feasible solution of smaller cost.

Definition 1. *Given a collection $\mathcal{P}$ of paths on the plane and a vertical line at point $x_0 \in \mathbb{R}$, the **shadow** at x_0 is the number of legs of the paths in $\mathcal{P}$ that have an intersection with the vertical line at x_0. The shadow of a given range $[a, b]$ is defined to be the maximum shadow of values $x_0 \in [a, b]$.*

Note that if a solution is self-crossing, the operation of uncrossing (which reduces the cost) does not increase the shadow

Suppose the sequence of points of OPT is $p_1, p_2, \ldots, p_\sigma$ and the straight lines connecting these points (i.e. legs of OPT) are $\ell_1, \ell_2, \ldots, \ell_\sigma$ where ℓ_i connects two points p_i, p_{i+1} (with $p_{\sigma+1} = p_1$), and each s_i has at least one point p_j on it. We consider OPT oriented in this order, i.e. going from p_i to p_{i+1}. Since all segments have distinct x-coordinates, we can assume no two consecutive points p_i, p_{i+1} are on the same line segment by short-cutting (so no leg ℓ_i is vertical) and all points p_i on distinct line segments have distinct x-coordinates.

As mentioned before, let the length of the sides of the minimal bounding box of an instance of the problem be $L \times H$. The proof of the following theorem, which has the same setting as our toy example in the introduction, appears in Subsection 4.3, where we use some of the definitions and lemmas given throughout this section.

Theorem 3. *If $H \leq 3$, then the shadow of an optimum solution is at most 2.*

By proving this special case, we show we can instead focus on the cases that H is large. So from now on, we assume that $H > 3$. Our first main goal is to prove Theorem 2.

Definition 2. *Given a segment s and a leg ℓ incident to a point on s, we say ℓ is to the **left** of s if ℓ is entirely in the subplane $x \leq x(s)$; and ℓ is to the **right** of s if ℓ is entirely in the subplane $x \geq x(s)$.*

Since there are no vertical legs, there is no leg that is both to the left and to the right of a segment of the instance at the same time. Consider any segment s_i and suppose that ℓ_j, ℓ_{j+1} are the two legs of OPT with common end-point p_j that is on s_i. Let s_i^t and s_i^b denote the top and the bottom tips of s_i. We consider 3 possible cases for the location of p_j and the arrangement of ℓ_j, ℓ_{j+1}. Informally, one possibility is that the

two legs ℓ_j, ℓ_{j+1} form a straight line that crosses s_i at p_j; one possibility is that the two legs are touching s_i at one of its tips (i.e. $p_j = s_i^t$ or $p_j = s_i^b$) such that one is to the left and one is to the right of s_i and they don't make a straight line, and the third possibility is that the two legs ℓ_j, ℓ_{j+1} are on the same side (both left or both right) of s_i.

Observation 1. *Consider any segment s_i (with top/bottom points s_i^t,s_i^b) and suppose that ℓ_j, ℓ_{j+1} are the two legs of OPT with common end-point p_j that is on s_i. Then either:*

- *The subpath of OPT going through p_{j-1}, p_j, p_{j+1} forms a straight line (i.e. $\angle \ell_j p_j \ell_{j+1} = \pi$), and ℓ_j, ℓ_{j+1} are on two sides (left/right) of s_i; then we call p_j a **straight point**, or*
- *p_j is a tip of s_i (i.e. $p_j = s_i^t$ or $p_j = s_i^b$), $\angle \ell_j p_j \ell_{j+1} \neq \pi$ and ℓ_j and ℓ_{j+1} are on two sides of s_i (one left and one right); in this case p_j is called a **break point**, or*
- *both ℓ_j, ℓ_{j+1} are on the left or on the right of s_i; in this case p_j is called a **reflection point**.*

For the case of a reflection point p_j with two legs ℓ_j, ℓ_{j+1}, if both legs are to the left of the segment it is called a *left reflection* point and otherwise it is a *right reflection* point. Also note that if ℓ_j, ℓ_{j+1} are on the two sides of s_i and $\angle \ell_i p_j \ell_{i+1} \neq \pi$, then p_j must be a tip, or else we could move p_j slightly up or down and reduce the length of OPT (see Figure 4).

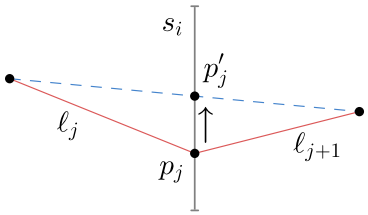


FIGURE 4. If p_j isn't a tip of s_i, then ℓ_j, ℓ_{j+1} must be collinear

Lemma 1. *If P is a subpath of OPT with end-points p, q where both are to the right of a vertical line Γ, and if P crosses Γ, then the left-most point on P to the left of Γ is a right reflection point (symmetric statement holds for opposite directions).*

Proof. Let r (on segment s) be the left-most point P visits, so both subpaths $P_{pr} = p \to r$ and $P_{qr} = q \to r$ are entirely to the right of r, in particular the two legs ℓ^- and ℓ^+ of P incident to r (which are the last two legs of the subpaths P_{pr}, P_{qr}) must be on the right of s which implies that r is a right reflection point. □

Definition 3. *Consider an arbitrary reflection point r on a segment s. Let the two legs of OPT incident to r visited before and after r (on the orientation of OPT) be ℓ^- and ℓ^+, respectively. ℓ^- is said to be **on top of** ℓ^+ if all the points of ℓ^- have larger y-coordinate than all of the points of ℓ^+. In this case we also call ℓ^- the **upper leg** and ℓ^+ the **lower leg**. Also, in this case, r is called a **descending** reflection point. If ℓ^+ is on top of ℓ^-, then r is called an **ascending** reflection point.*

Definition 4. *If ℓ_j, ℓ_{j+1} are two legs incident to a reflection point p on a segment s, if the angle between ℓ_j and s is the same as the angle between ℓ_{j+1} and s (i.e. ℓ_{j+1} is like the reflection of a ray ℓ_j on mirror s) then p is called a **pure reflection** point.*

Lemma 2. *Any reflection point that is not a tip of a segment is a pure reflection point.*

Proof. Suppose p_j is a reflection point on s_i and is not a tip of it. If the two legs ℓ_j, ℓ_{j+1} don't have the same angle with s_i, then we can move p_j along s_i slightly up or down and one of the moves will decrease the cost of OPT, a contradiction. □

We say a subpath contains a reflection point p_j if p_j is not the start or end vertex of the subpath (i.e. both legs of incident to p_j belong to that subpath.)

Lemma 3. *If a sweeping vertical line Γ moves left to right on the x-axis, the only values of x for which the shadow at Γ changes will be when Γ hits a reflection point on that x-coordinate. Specifically, this means that any subpath of OPT that doesn't contain a reflection point, must have a shadow of 1 throughout its length.*

Proof. According to Observation 1, we can see that straight points or break points will always contribute 1 to the shadow of Γ. But reflection points, depending on which direction the sweeping line moves, will either

increase or decrease the shadow by 2. If a path doesn't contain any reflections, it means that it can only contain straight points or break points, meaning its shadow throughout its length will be equal to 1. □

Definition 5. *Let P_1 and P_2 be any two subpaths of OPT. We say P_1 is **above** P_2 in range $I = [x_0, x_1]$ if for every vertical line Γ with $x(\Gamma) \in I$, the top-most intersection of Γ with these two paths is a point on P_1. We say P_2 is **below** P_1 if the bottom-most intersection of Γ with P_1, P_2 is a point on P_2. Similarly, we say L_1 is to the **left** of L_2 in range $I' = [y_0, y_1]$ if for every horizontal line Λ with $y(\Lambda) \in I'$, the left-most intersection point of Λ with L_1, L_2 (i.e. one with the least x value) always belongs to L_1. We say L_2 is to the **right** of L_1 if the right-most intersection of Λ is with L_2.*

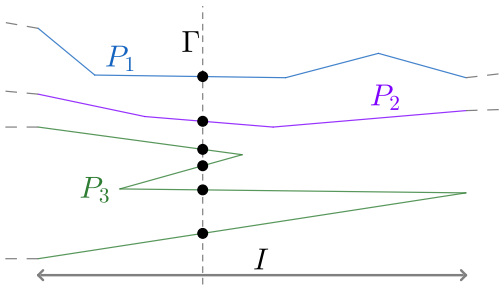


FIGURE 5. In range I, P_1 is above P_2, P_3, and P_2 is above P_3

Lemma 4. *For any distinct points p_j and $p_{j'}$ on OPT, following OPT according to its orientation, either the path from p_j to $p_{j'}$ or the path from $p_{j'}$ to p_j must contain at least one reflection point.*

Proof. Without loss of generality, assume $x(p_j) < x(p_{j'})$, and following the orientation of OPT starting from p_j, suppose the path from p_j to $p_{j'}$ does not contain any reflection points (or the statement of lemma holds). According to Observation 1, the x-coordinate of points on OPT will not decrease if and only if the path contains only straight points or break points. The path from $p_{j'}$ to p_j has to have a decrease in the x-coordinate, due to $x(p_{j'}) > x(p_j)$, which is only possible if there is a reflection in this part of the path. □

Lemma 5. *Let r_j be any reflection point on OPT, say it is a right reflection point, with incident legs ℓ_i, ℓ_{i+1}. Without loss of generality, assume that ℓ_i is above ℓ_{i+1}. Take any two subpaths P_1 and P_2 of OPT both starting at r_j with shadow of 1 such that $\ell_i \in P_1$ and $\ell_{i+1} \in P_2$. If there is a vertical line Γ with $x(\Gamma) > x(r_j)$ crossing both P_1 and P_2, then P_1 will be above P_2 in range $I = [x(r_j), x(\Gamma)]$.*

Proof. Note that for any vertical line Γ' with $x(\Gamma') \in I$, both P_1 and P_2 will intersect with it. Now assume the contrary, that P_1 is not above P_2. This means for some vertical line Γ' with $x(\Gamma') \in I$, there are points p_1 and p_2 on Γ' such that $p_1 \in P_1$, $p_2 \in P_2$, and $y(p_2) > y(p_1)$. Since both P_1 and P_2 have a shadow of 1, then using Lemma 3, we get that neither of them have a reflection point; this implies that the value of the x-coordinate on both P_1 and P_2 is monotone (or else there must be a reflection point). Since P_1 travels from r_j to p_1 and P_2 travels from r_j to p_2, both are crossing the same vertical lines (at $x = x(r_j)$ and Γ'). Now, because ℓ_i is above ℓ_{i+1} but p_1 is below p_2, we conclude that P_1 and P_2 will intersect with each other in the area between the vertical lines Γ' and $x = x(r_j)$. This is a contradiction, hence the lemma. □

Lemma 6. *Among the set of points visited by OPT following its orientation, suppose $p_j, p_{j'}$, $j < j'$ (on segments s_i, $s_{i'}$, respectively) are two consecutive reflection points (i.e. no other reflection point exists in between them). Then p_j and $p_{j'}$ cannot be both left or both right reflection points. Furthermore, if s_i is to the left of $s_{i'}$ then p_j is a right reflection and $p_{j'}$ is a left reflection (the opposite holds if $s_{i'}$ is to the left of s_i).*

Proof. Without loss of generality, assume that s_i is to the left of $s_{i'}$, meaning $x(p_j) < x(p_{j'})$. By way of contradiction, first suppose both p_j and $p_{j'}$ are right reflection points, i.e. the two legs incident to p_j (ℓ_j, ℓ_{j+1}) and the two legs incident to $p_{j'}$ $(\ell_{j'}, \ell_{j'+1})$ are on the right of s_i and right of $s_{i'}$, respectively. This means following the orientation on OPT, along ℓ_j we have a decrease in x-coordinate, then following ℓ_{j+1} have an increase, then again following $\ell_{j'}$ have a decrease and following $\ell_{j'+1}$ have an increase. So the value of the x-coordinate isn't monotone in the subpath of OPT from p_j to $p_{j'}$ (excluding these two points themselves),

because the legs ℓ_{j+1} and $\ell_{j'}$ are visited in this path in this order. Similar to the proof in Lemma 4, we see that this is only possible if there is a reflection point on this subpath, which contradicts the assumption that $p_j, p_{j'}$ are consecutive. Similar argument implies that we cannot have both $p_j, p_{j'}$ being left reflections or p_j being a left reflection and $p_{j'}$ being a right reflection; otherwise, the leg after visiting p_j will have decreasing x-value while it will have to visit $p_{j'}$ eventually, which has a larger x-value. So the path from p_j to $p_{j'}$ must include another reflection point, again a contradiction. □

Corollary 1. *Consecutive reflection points in OPT alternate between left reflections and right reflections.*

Lemma 7. *If a segment s_i has a reflection point p_j, then it cannot have any other intersections with OPT (i.e. no other point p'_j can be on s_i).*

Proof. Assume otherwise, that a segment s_i contains a reflection point p_j with legs ℓ_j and ℓ_{j+1}, and another point $p_{j'}$ on s_i. We can by-pass p_j locally and reduce the length of OPT which would be a contradiction. More specifically, let $R^- \in \ell_j$ and $R^+ \in \ell_{j+1}$ be points on the legs that have a distance of $\delta > 0$ from p_j. By replacing the subpath $R^- \to p_j \to R^+$ with $R^- \to R^+$, the total cost of OPT will decrease, which gives us a contradiction. □

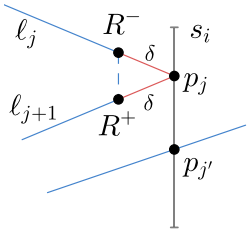


FIGURE 6. There can't be another $p_{j'} \in s_i$ if $p_j \in s_i$ is a reflection

We decompose the problem into horizontal *strips* by drawing some parallel horizontal lines, which we call *cover-lines*. Starting from the bottom tip of the top-most segment, draw parallel horizontal lines that are 1-unit apart, these are our *cover-lines*. Each input segment is considered "covered" by the top-most (i.e. the first) cover-line that intersects it. Let's label these cover-lines by $C_1, C_2, \ldots$.

Definition 6 (strip, top/bottom segments)**.** *The region of the plane between two consecutive cover-lines $C_\tau, C_{\tau+1}$ is called a **strip** and denoted by S_τ. We consider $C_\tau, C_{\tau+1}$ part of S_τ as well. The input line segments that are intersecting the top cover-line of S_τ (C_τ) are called **top segments,** and the segments covered by the bottom cover-line ($C_{\tau+1}$) are called **bottom segments** of the strip.*

We show the near-optimum solution guaranteed by Theorem 2 has more structural properties that will be defined later. Note that once we prove that theorem, it follows that if we restrict a solution to $h > 1$ many strips, then the shadow is bounded by $O(h/\varepsilon)$ as well.

For now, let us focus on an (arbitrary) strip S_τ and imagine we cut the plane along $C_\tau, C_{\tau+1}$ and look at the pieces of line segments of the instance left inside this strip, along with pieces of OPT inside S_τ. Each top segment is now a partial segment in S_τ that has one end on C_τ; and each bottom segment has one end on $C_{\tau+1}$. Let OPT_τ be the restriction of OPT to S_τ. For each leg of OPT that intersects C_τ or $C_{\tau+1}$, we add a dummy point at the intersection(s) of that leg with C_τ and $C_{\tau+1}$ (so that the components of OPT_τ become consistent with our definition of legs). So OPT_τ can be seen as a collection of subpaths within S_τ (possibly along C_τ or $C_{\tau+1}$). Following the orientation of OPT, each subpath of OPT_τ is formed by it intersecting with S_τ, traveling within S_τ (possibly along one of the cover-lines), and then exiting S_τ. Using the dummy points added, each path in OPT_τ is a subpath of OPT that is between two points on cover-lines (these are called the entry points of the path with the strip. A formal definition is provided below).

Recall Definition 5 of paths being above or below each other. Having the definition of top/bottom segments, we get the following:

Observation 2. *Consider OPT_τ, the restriction of OPT to any strip S_τ. Take any two subpaths of OPT_τ like P_1 and P_2 such that P_1 is above P_2 in some range I. If s_t is any top segment in range I that P_2 intersects with, then P_1 will also intersect with it. Similar statement holds for bottom segments if P_2 is below P_1.*

Definition 7 (entry points, loops, ladders)**.** *For each subpath P_j of OPT_τ, let e_j and o_j be the first and last intersections of P_j with the interior of S_τ. Points e_j and o_j are called the* ***entry points*** *of P_j.*
If both e_j and o_j lie on the same cover-line (either C_τ or $C_{\tau+1}$), then P_j is called a ***loop****, otherwise it's called a* ***ladder****. If a subpath of OPT_τ enters S_τ at e_j on a cover-line and follows on that cover-line to point o_j and exits the strip, it is a special case of loop that we refer to as a* ***cover-line loop****.*

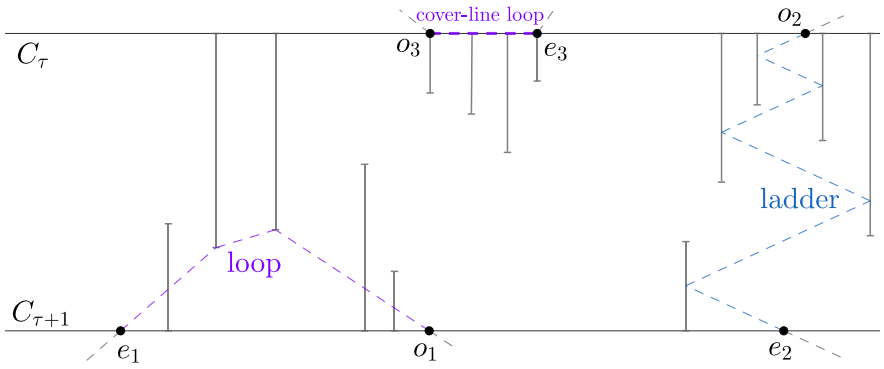


FIGURE 7. An example of loops and ladders in a strip S_τ (i.e. area between cover-lines $C_\tau, C_{\tau+1}$)

Since we're assuming $H > 3$ (see Theorem 3), we get that OPT is not limited to a single strip, and that it has to indeed enter and exit any given strip that it intersects with (i.e. there is no strip that OPT completely lies inside it).

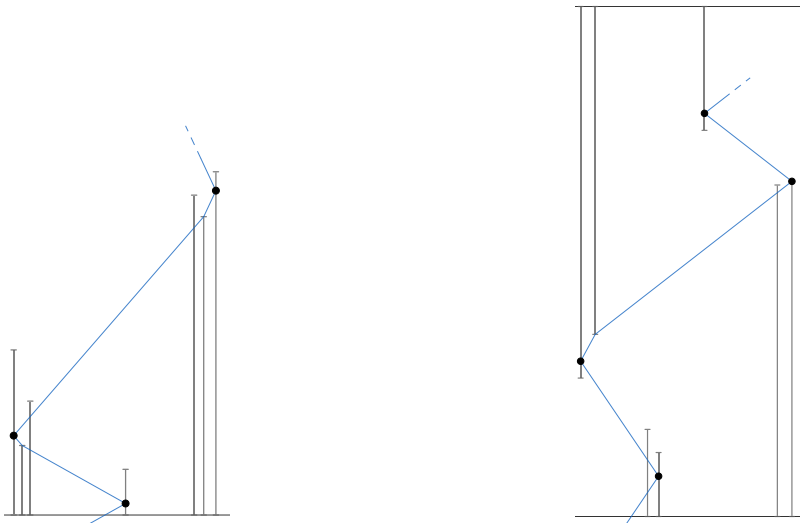
Note that if a path of OPT_τ is a cover-line loop, i.e. a section of the line C_τ or $C_{\tau+1}$, then the entry points of that path must be the two end-points of this section. In other words, if for a cover-line loop of OPT_τ the first point is e_j on (say) C_τ, and the last point is o_j on C_τ, then this subpath must be travelling straight from e_j to o_j without any change of direction. This is true because otherwise, that cover-line loop would have to go back and forth on some portion on a cover-line, which is only possible if it's self-intersecting; but this is against our assumption that OPT is not self-crossing.

The two structures defined below (called a zig-zag and a sink) are the two configurations that can cause a large shadow.

Definition 8 (Zig-zag/Sink)**.** *Consider any loop or ladder of OPT_τ, call it P. Let $\mathcal{R} = r_1, r_2, \ldots, r_t$ be the sequence of points of P that are reflection points (indexed by the order they're visited). Consider any maximal sub-sequence $r_j, r_{j+1}, \ldots, r_q$ of $\mathcal{R}$ (with $q \geq j+1$) such that all are ascending or all are descending reflections, and the segments containing them alternate between top and bottom segments; then the subpath P that starts at r_j and ends at r_q is called a* ***zig-zag****.*
If $r_j, r_{j+1}, \ldots, r_q$ is a maximal sub-sequence of $\mathcal{R}$ that are all ascending or all descending, and all belong to top segments or all belong to bottom segments; then the subpath P that starts at r_j and ends at r_q is called a ***sink*** *(see Figure 8).*

Note that each zig-zag has at least two reflection points (or else it is a sink instead). Also, using Corollary 1, the reflection points in a zig-zag or sink alternate between left and right reflections. The next lemma is used critically to show that very specific structures (made by zig-zags and sinks) are responsible for having a large shadow along a ladder or loop in OPT_τ. Additionally, we can partition each ladder/loop into parts (subpaths), such that the shadow of the ladder/loop is equal to the maximum shadow among these parts, and each part is a path that consists of up to three sinks and/or zig-zags. So the shadow of a loop/ladder is within $O(1)$ of the maximum shadow of zig-zag/sinks along it.

Lemma 8 (First major lemma). *Consider any strip S_τ and any ladder or loop $P \in OPT_\tau$ within S_τ. Suppose the sequence of reflection points of P is $r_1, \ldots, r_q$. These reflection points can be partitioned into disjoint parts, say part i consists of reflection points $r_{a_i}, r_{a_i+1}, \ldots, r_{a_j}$, where the subpath of P from r_{a_i} to r_{a_j} is concatenation of up to three sections in the following order:* ***a)*** *A sink , followed by* ***b)*** *A zig-zag , followed by* ***c)*** *A sink, where any of these three sections can possibly be empty, and the last reflection of a section is common with the first reflection of the next section. Furthermore, for any vertical line Γ, there is*



(A) A sink (B) A zig-zag

FIGURE 8. Examples of sinks and zig-zags. The bold black dots represent the reflection points along these paths

at most one of these parts (of the partition) that intersects with it, i.e. the shadow of the ladder/loop is the maximum shadow among the parts plus 2.

The proof of this lemma is rather involved and appears in Subsection 4.1. To give an idea of the proof, we essentially show that for any loop or ladder in any strip, the vertical line at which the largest shadow for that loop or ladder happens, can intersect with at most two sinks and a zig-zag. So the shadow of a loop or ladder is within $O(1)$ of the maximum shadow of the zig-zags and sinks along that.

2.2. **Properties of a Near-Optimum Solution.** In this section we introduce some lemmas that describe structures of a near-optimum solution with small shadow. The next three major lemmas, namely Lemmas 9, 10, and 11 are the main lemmas used in the proof of Theorem 2. We prove these in Subsections 4.2.1, 4.2.2, and 4.2.3, respectively. We make alterations to an assumed optimum solution such that some new structural properties hold; we ensure that the alterations have a limited additional cost.

Lemma 9 (**Second major lemma**). *Consider OPT_τ for an arbitrary strip S_τ and let opt_τ be the total cost of OPT_τ. Given any $\varepsilon > 0$, we can change OPT_τ to a solution of cost at most $(1+O(\varepsilon))opt_\tau$ where the shadow of each zig-zag and sink is at most $O(1/\varepsilon)$.*

The proof of this lemma appears in Subsection 4.2.1. The following corollary immediately follows from Lemmas 8 and 9:

Corollary 2. *There is a $(1+\varepsilon)$-approximate solution in which all loops/ladders have shadow $O(1/\varepsilon)$.*

Definition 9. *Let $\mathcal{R} = p_i, p_{i+1}, \ldots, p_q$ be any sequence of consecutive points in OPT such that p_i and p_q are reflection points. If none of p_j's $(i < j < q)$ in $\mathcal{R}$ is a tip of a segment, then $\mathcal{R}$ is called a* ***pure reflection sequence****.*

So each point in $\mathcal{R}$ is either a straight point or a pure reflection according to Lemma 2. The following lemma (whose proof appears in Section 4.2.2) shows the existence of a near-optimum with bounded size pure reflection sequences.

Lemma 10 (**Third major lemma**). *Consider OPT_τ for an arbitrary strip S_τ and suppose the total length of legs of OPT_τ is opt_τ. Given $\varepsilon > 0$, we can change OPT_τ to a solution of cost at most $(1+\varepsilon)opt_\tau$ in which the size of any pure reflection sequence is bounded by $O(\frac{1}{\epsilon})$.*

Our next goal is to show that for any vertical line, it can intersect at most $O(1)$ many loops or ladders of OPT_τ in a strip S_τ. This together with the above corollary implies there is a $(1+\varepsilon)$-approximate solution s.t. the shadow in each strip S_τ is $O(1/\varepsilon)$.

Definition 10. *A collection of loops and or ladders are said to be **overlapping** with each other if there is a vertical line that intersects all of them.*

Lemma 11 (**Fourth major lemma**). *Consider OPT_τ, the restriction of OPT to any strip S_τ. We can modify the solution (without increasing the shadow or the cost) such that there are at most $O(1)$ loops or ladders in OPT_τ that all are overlapping with each other.*

We prove this Lemma in Section 4.2.3 as it consists of multiple parts.

2.3. **Proof of Theorem 2.** Assuming the correctness of main lemmas defined in this section (i.e. Lemmas 8, 9, 10, and 11), we can now prove Theorem 2.

If the height of the bounding box is at most 3, refer to Theorem 3. Otherwise, consider any strip S_τ (to be more precise, S_τ can be any arbitrary strip of height 1 in the plane). Using Lemma 9 for parameter $\varepsilon_1 = \frac{\varepsilon}{2}$, there is a solution $\mathcal{O}''$ of cost at most $(1+\frac{\varepsilon}{2})\cdot \text{opt}$ where the shadow of each sink and zig-zag is bounded by $O(\frac{1}{\varepsilon/2}) = O(1/\varepsilon)$. By Lemma 8, each loop or ladder in S_τ has a shadow that is at most 3 times the maximum shadow of a sink or zig-zag in it, plus two. So each loop or ladder has shadow $O(1/\varepsilon)$. Finally, Lemma 11 shows that there can be at most $O(1)$ overlapping loops or ladders in a strip. Thus, the overall shadow of $\mathcal{O}''$ in S_τ is bounded by $O(1/\varepsilon)$. Furthermore, we apply Lemma 10 on O'' for parameter $\varepsilon_2 = \frac{\varepsilon}{\varepsilon+2}$ to get a solution $\mathcal{O}'$. This new solution has the property that with an additional cost of factor $(1+\frac{\varepsilon}{\varepsilon+2})$ compared to $\mathcal{O}''$, the size of any pure reflection sequence is bounded by $O(\frac{\varepsilon+2}{\varepsilon}) = O(1/\varepsilon)$. The total cost of $\mathcal{O}'$ is at most

$$\begin{aligned}(1+\varepsilon_1)\cdot(1+\varepsilon_2)\cdot\text{opt} &= (1+\tfrac{\varepsilon}{2})\cdot(1+\tfrac{\varepsilon}{\varepsilon+2})\cdot\text{opt}\\ &= (1+\tfrac{\varepsilon}{2}+\tfrac{\varepsilon}{\varepsilon+2}+\tfrac{\varepsilon^2}{2(\varepsilon+2)})\cdot\text{opt}\\ &= (1+\varepsilon(\tfrac{1}{2}+\tfrac{1}{\varepsilon+2}+\tfrac{\varepsilon}{2(\varepsilon+2)}))\cdot\text{opt}\\ &= (1+\varepsilon)\cdot\text{opt},\end{aligned}$$

resulting in the statement of the theorem. □

Note: Although having a bound on the length of pure reflection sequences is not in the statement of the theorem, we use this extra property crucially in designing our DP to find a near-optimum solution.

3. Main Algorithm and Reduction to Structured Bounded Height Instances

As mentioned in the introduction, we follow the paradigm of Arora Arora (1998) for designing a PTAS for classic Euclidean TSP with some modifications. We focus more on defining the modifications that we need to make to that algorithm.

First, we describe the main algorithm and how it reduces the problem into a collection of instances with a constant-height bounding box. We show how those instances can be solved using another DP (referred to as the *inner DP*), and how we can combine the solutions for them using another DP (referred to as the *outer DP*) to find a near-optimum solution of the original instance. Recall that in Subsection 1.3, we assumed the minimal bounding box of the instance has length L and height H and we defined $B = \max\{L, H-2\}$, and also we can assume that $B \le \frac{n}{\varepsilon}$. We moved each line segment to be aligned with a grid point with side length $\frac{\epsilon B}{n^2}$ (at a loss of $(1+\varepsilon)$ at approximation). Now, we scale the grid (as well as the line segments of the instance) by a factor of $\rho = \frac{4n^2}{\epsilon B}$ so that each grid cell has size 4. We obtain an instance where each line segment has length ρ, all have even integer coordinates, any two segments are at least 4 units apart, and the bounding box has size $N = O(n^2/\varepsilon)$. Let this new instance be $\mathcal{I}$. Note that if we define cover-lines as before but with a spacing of ρ, all the arguments for the existence of a near-optimum solution with a bounded shadow in any strip (the area between two consecutive cover-lines) still hold. We will present a PTAS for this instance. It can be seen that this implies a PTAS for the original instance of the problem. From now on, we use OPT to refer to an optimum solution of instance $\mathcal{I}$, and opt to refer to its cost. Note that since the bounding box has side length N, then opt $\ge 2N$.

3.1. **Dissecting the Original Instance into Smaller Subproblems.** Similar to Arora's approach, we do the hierarchical dissectioning of the instance into nested squares using random axis-parallel dissectioning lines, and put portals at these dissecting lines. We continue this dissectioning process until the distances between horizontal (and so vertical) dissecting lines is $h\cdot\rho$ for $h = \lceil 1/\varepsilon \rceil$. So at the leaf nodes of our

recursive decomposition quad-tree, each square is $(h \cdot \rho) \times (h \cdot \rho)$, and the height of the decomposition is $\log(N/\rho h) = O(\log n)$ since $B \leq \frac{n}{\varepsilon}$. We choose vertical dissecting lines only at odd x-coordinates so no line segment of the instance will be on a vertical dissecting line.

We define our cover-lines C_τ based on these horizontal dissecting lines carefully. Consider the first (horizontal) dissecting line we choose, this will be a cover-line, and then moving in both up and down directions from this line, we draw horizontal lines that are ρ apart. These will be all the cover-lines. Label the cover-lines from the top to bottom by $C_1, C_2, \ldots, C_\sigma$ in that order. As before, the smallest index τ such that C_τ crosses a line segment is the cover-line that "covers" that line segment.

We partition the cover-lines into h groups based on their indices: Group G_j contains all those cover-lines with index τ where $j = \tau \pmod h$. Let G_{j*} be the group of cover-lines that includes the first horizontal dissecting line, and hence all the other horizontal dissecting lines as well. The arguments for the case of unit-length line segments to show there is a near-optimum solution in which the shadow in each strip of height 1 is $O(1/\varepsilon)$ (Theorem 2), also imply the same for the scaled instance $\mathcal{I}$. Furthermore, if we consider h consecutive strips, i.e. the area between two consecutive cover-lines in the same group G_j, then there is a near-optimum solution that has shadow $O(h/\varepsilon) = O(1/\varepsilon^2)$. Our goal is to show that, at a $(1+\varepsilon)$-factor loss, we can simply drop the line segments that are intersecting the horizontal dissecting lines (i.e. all those intersecting cover-lines in G_{j^*}) with appropriate consideration of portals (to be described). Removing the line segments that cross the dissecting lines allows us to decompose the instance into "independent" sub-instances that interact only via portals.

For each cover-line C_τ, we define a set B_τ of disjoint (horizontal) *intervals* of length ρ placed on it so that each line segment covered by C_τ, is intersecting one of these intervals. To define B_τ, move on C_τ, from left to right, start by placing the left corner of the first interval of B_τ on it at the intersection of the left-most segment covered by C_τ; all the segments covered by C_τ intersecting this interval are considered "covered" by this interval. Next, pick the first segment to the right of the latest interval that is intersecting C_τ, but not intersecting (and so not covered by) the previous intervals, and place the left point of the next interval of B_τ at that intersection (all the segments intersecting C_τ and this interval are now covered by this interval). Continue this process until all segments on C_τ are covered by an interval (see Figure 9). Let $\mathcal{B} = \cup_{\tau=1}^{\sigma} B_\tau$.

Observation 3. *A segment covered by an interval of cover-line C_τ and another segment covered by an interval of cover-line $C_{\tau+2}$ are at least ρ apart ($\tau \leq \sigma - 2$).*

Lemma 12. $opt \geq \dfrac{\rho \cdot |\mathcal{B}|}{6}$.

Proof. For each B_τ, let $i_1, i_2, \ldots, i_\eta$ be the intervals on C_τ ordered from left to right. Now partition B_τ into $O_\tau \cup E_\tau$ where O_τ consists of intervals i_q with an odd q, and E_τ consists of those with even q's. We also partition C_τ's into 3 groups based on the value of $\tau \pmod 3$. We get a partition of all intervals into 6 groups based on: Whether an interval on C_τ is in O_τ or E_τ (two choices), and what $\tau \pmod 3$ is (three choices). Let N_j's $(1 \leq j \leq 6)$ be the total number of intervals in these 6 parts. Note that $\sum_{j=1}^{6} N_j = |\mathcal{B}|$, and any two segments s, s' covered by intervals from different groups are at least ρ apart (if they there are covered by intervals in the same cover-line then they are ρ apart horizontally and if they covered by intervals in different cover-lines then by Observation 3 they are at least ρ apart). So an optimum solution for the instance that only contains segments covered by intervals of part N_j, must have cost at least ρN_j as it must have at least N_j legs of size at least ρ. Since one of these parts has size at least $|\mathcal{B}|/6$, the statement follows. □

Lemma 13. *For a j chosen randomly from $[1..h]$, we have*

$$\mathbb{E}[\ \rho \sum_{C_\tau \in G_j} |B_\tau|\ ] = O(\varepsilon \cdot opt).$$

Proof. For each $1 \leq j \leq h$, let $\mathcal{B}_j = \bigcup_{C_\tau \in G_j} B_\tau$. Using Lemma 12, we have $\sum_{j=1}^{h} |\mathcal{B}_j| = |\mathcal{B}| \leq 6 \cdot opt/\rho$. Now we obtain

$$\begin{aligned}\mathbb{E}[\rho \sum_{C_\tau \in G_j} |B_\tau|] &= \rho \cdot \mathbb{E}[\sum_{C_\tau \in G_j} |B_\tau|] \\ &= \rho \cdot \mathbb{E}[|\mathcal{B}_j|] \\ &= \frac{\rho}{h} \cdot |\mathcal{B}| \\ &\leq \frac{\rho}{h} \cdot \frac{6 \cdot opt}{\rho} = O(\text{opt}/h) = O(\varepsilon \cdot \text{opt}).\end{aligned}$$

□

Similar to Arora's scheme for TSP, for $m = O(\frac{1}{\varepsilon} \log(N/\rho h))$, we place portals at all 4 corners of a square in the decomposition, plus an additional $m-1$ equally distanced portals along each side (so a total of $4m$ portals on the perimeter of a square of the dissection). For simplicity, we assume m is a power of 2 and at least $\frac{4}{\varepsilon} \log(N/\rho h)$. We say a tour is *portal respecting* if it crosses between two squares in our decomposition only via portals of the squares. A tour is *r-light* if it crosses the portals on each side of a square of the dissection at most r times.

For classic (point) TSP, it can be shown that there is a near-optimum solution that is portal respecting and r-light for $r = O(1/\varepsilon)$. Our goal is to show a similar statement, except that we want the restriction of the tour to each "base" square of side length $O(h \cdot \rho)$ to have bounded (by $O(h/\varepsilon) = O(1/\varepsilon^2)$) shadow as well. We then show that we can find an optimum solution with a bounded shadow for the base cases using a DP. This will be our *inner DP*. We then show how the solutions of for the 4 sub-squares of a square in our decomposition can be combined into a solution for the bigger subproblem using the outer DP.

We show that at a small loss in approximation (i.e. $O(\varepsilon \cdot \text{opt})$), we can drop all the line segments of input that are intersecting the horizontal dissecting lines (i.e. covered by a cover-line in group G_{j^*}), solve appropriate subproblems, and then extend the solutions to cover those dropped segments. This modification requires certain portals of each square in the decomposition to be visited in the solution for that square.

We show there is a feasible solution that visits all the remaining segments as well as the "required" portals, of total cost at most $(1+\varepsilon) \cdot \text{opt}$, and that such a solution can be extended to a feasible solution visiting all the segments of the original instance (i.e. including the ones that we dropped) at an extra cost of $O(\varepsilon \cdot \text{opt})$.

3.1.1. *Dropping the segments intersecting horizontal dissecting lines.* This Subsection is dedicated to proving the following lemma:

Lemma 14. *Given instance $\mathcal{I}$, there is another instance $\mathcal{I}'$ that is obtained by removing all the segments that are crossing cover-lines in G_{j^*} (i.e. intersecting horizontal dissecting lines), and instead some of the portals around (more precisely, the top and bottom sides of) each square of quad-tree dissection are required to be covered (visited); such that there is a solution for $\mathcal{I}'$ of cost at most $(1+O(\varepsilon)) \cdot opt$, and such a solution can be extended to a feasible solution of $\mathcal{I}$ of cost at most $(1+O(\varepsilon)) \cdot opt$. Furthermore, the shadow of the solution for $\mathcal{I}'$ between any two consecutive cover-lines in G_{j^*} is at most 4 more than the shadow of OPT between those two lines.*

We say the edges of the bounding box are *level 0* dissecting lines, the first pair of dissecting lines are *level 1* dissecting lines, and so on. Consider a square S in our hierarchical decomposition and suppose it is cut into four squares S_1, S_2, S_3, S_4 by two dissecting lines where the horizontal one, line Γ, is the cover-line C_τ from G_{j^*}, and is a level j dissecting line. Recall that we place a total of $2m$ portals along Γ inside S; m portals on the common sides of S_1, S_4 and m along the common side of S_2, S_3. Define $B_\tau(S)$ to be the set of intervals in B_τ (intervals of C_τ) that cover a segment that lies inside S (and so intersects with Γ) (see Figure 9).

For each $b \in B_\tau(S)$, suppose $p(b)$ is the nearest portal to it in S among the portals on Γ, and let $s(b)$ be the left-most segment covered by b that is in S. We are going to modify OPT in the following way: Consider a point p_s on $s(b)$ visited by OPT. Insert the following "legs" to the path: travel from p_s vertically along $s(b)$ until you arrive at its intersection with Γ, i.e. arrive at interval b (this length is at most ρ), then travel along Γ to the right-most segment covered by b (this is also at most ρ), and then travel to $p(b)$, and then travel back to p_s. For every other segment s' covered by b in S, we are going to short-cut any point on s'

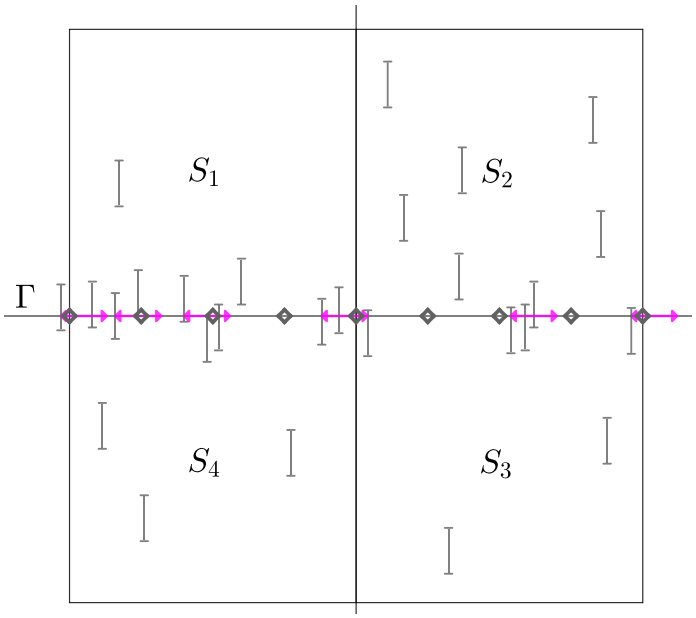


FIGURE 9. Breaking a square S into 4 smaller squares. The magenta parts on line Γ (i.e. the cover-line C_τ) show the interval set $B_\tau(S)$

that was visited by OPT as all these segments are now covered by the newly added legs (see Figure 10). We also short-cut the second visit to p_s.

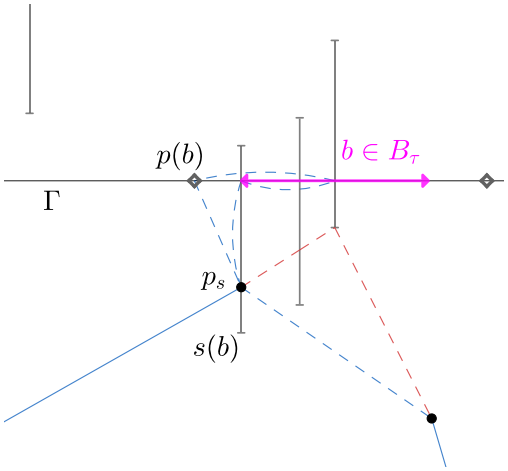


FIGURE 10. The modified solution for dropping segments crossing horizontal dissecting lines: follow the blue dashed lines from p_s; red dashed lines are the discarded parts of the original path

Using triangle inequality, the expected length of the new legs will increase the cost of the solution by at most $2\rho + 2||p_s p(b)|| \leq 2(\rho + \frac{N}{2^j m})$. We do this for all the intervals on Γ and inside S, i.e. if OPT visits a segment covered by that interval b, we change OPT to make a detour to visit $p(b)$ as well. Note that each interval $b \in B_\tau$ can belong to at most two $B_\tau(S)$'s (two adjacent squares that b intersects with), and the intervals for which this modification can happen for, are at least $h \cdot \rho$ apart because that is the minimum size of a square of the dissection.

Given the random choice of our dissecting lines, since dissecting lines are $h \cdot \rho$ apart, are randomly chosen, and each interval has length ρ, the probability that an interval $b \in B_\tau$ appears in two $B_\tau(S)$'s (i.e. cut by a dissecting line), is at most $1/h = \varepsilon$. Also, each cover-line in G_{j^*} is a level j dissecting line with probability

$2^{j-1}/(N/\rho h)$. Thus, the expected increase in the cost by this modification for all the interval of C_τ is at most

$$\begin{aligned}
&\sum_{j=1}^{\log(N/\rho h)} \Pr[\Gamma \text{ is level } j] \cdot (1+\varepsilon) \cdot |B_\tau| \cdot 2(\rho + \frac{N}{2^j m}) \\
&\leq 2(1+\varepsilon) \cdot |B_\tau| \cdot \sum_{j=1}^{\log(N/\rho h)} \frac{2^{j-1}}{N/\rho h} \cdot (\rho + \frac{N}{2^j m}) \\
&\leq 2(1+\varepsilon) \cdot |B_\tau| \cdot \frac{\rho h}{N} \cdot \left(\frac{N}{h} + \frac{N \log(N/\rho h)}{2m} \right) \\
&\leq (1+\varepsilon) \cdot |B_\tau| \cdot \rho \cdot (1+\varepsilon h) \\
&\leq 4\rho \cdot |B_\tau|.
\end{aligned}$$

Considering all cover-lines in G_{j^*}, this implies the total expected increase in the cost is at most $\sum_{C_\tau \in G_{j^*}} 4\rho|B_\tau|$, which combined with Lemma 13, implies with probability at least 1/2, the increase in total cost is at most $O(\varepsilon \cdot \text{opt})$. Each portal p that is visited by a detour as described above is called a *required portal*.

In fact, we can short-cut more paths so that the number of detours to each portal is bounded by 2. Informally, only the left-most interval to the left of p that has made a detour to p, along with the right-most interval to the right of p that has made a detour to p are sufficient to cover all the segments of the intervals in between them. More specifically, consider a portal p on Γ and let $b_L(p)$ be the left-most interval in $B_\tau(S)$ to the left of p that covered a segment whose path was detoured to visit p (null if there is no such interval). Similarly, let $b_R(p)$ be the right-most interval among $B_\tau(S)$ to the right of p that covered a segment whose path was detoured to visit p (this too can be null if there is no such interval). In other words, there was a segment $s_L = s(b_L(p))$ and a segment $s_R = s(b_R(p))$ that were visited by OPT, and we made a detour to p when OPT visited s_L and s_R. The detour from s_L to p covers all the segments of intervals between $b_L(p)$ and p. Similarly, the detour from s_R to p covers all the segments of interval between p and s_R. Thus, for any interval b' between $b_L(p)$ and $b_R(p)$, all the segments covered by b' are also covered by the detours of s_L and s_R. This means for all those intervals b', we can short-cut the segments covered by them entirely (in particular, they don't need to make a detour to p). Therefore, at most two intervals will have detours to p, namely $b_L(p)$ and $b_R(p)$. And the detours to different portals are disjoint, so the added detours don't overlap on Γ, and since short-cutting doesn't increase the shadow, we only add a shadow of at most 2 per cover-line to the solution. This implies that if we focus on the modified solution restricted to the strip between two cover-lines in G_{j^*}, it still has a bounded shadow. These arguments complete the proof of Lemma 14. □

3.2. **Dynamic Program.** As mentioned previously, the outer DP is responsible of combining the solutions of each "base-case" subproblem into a solution of the main instance of the problem; and the inner DP is responsible of solving each base-case (i.e. subproblems with a bounding box of size $O(1/\varepsilon)$).

3.2.1. *Outer DP.* The outer DP based on the quad-tree dissection is similar to the classic PTAS for Euclidean TSP. One can show that for $r = O(1/\varepsilon)$, there is an r-light portal respecting tour for $\mathcal{I}'$ with cost at most $(1+\varepsilon) \cdot \text{opt}'$, where opt$'$ is the cost of an optimum solution for $\mathcal{I}'$. The base case of this DP will be instances with bounding box of size $\rho \cdot h$. For such instances, we solve the problem using an inner DP that is described in the Subsection 3.2.2.

We will use the "patching lemma" the same way it is described in Arora's approach. We show there is a near-optimum solution for $\mathcal{I}'$ that is portal respecting and *r-light*, meaning each square in our quad-tree decomposition is crossed by the solution only r many times on each side for a parameter $r = O(1/\varepsilon)$. Then a DP similar to the point TSP (outer DP) will combine the solutions for the subproblems to find the solution for a bigger subproblem. Since we don't know which portals for each square are supposed to be "required" in $\mathcal{I}'$ (so that the solution can be extended to cover the dropped line segments), for each such square we "guess" the set of required portals in our DP; i.e. we will have an entry for each guessed set of portals on the horizontal sides of a square as the set of required portals in our DP. Since the number of portals is logarithmic, this guessing remains polynomially bounded. For now, assume that we know all the required portals, and hence, instance $\mathcal{I}'$ itself (even though $\mathcal{I}'$ is defined based on OPT which we don't know).

Consider instance $\mathcal{I}'$ and let OPT$'$ be the optimum solution for it, and let the cost of that solution be opt$'$. For each dissecting line Γ (vertical or horizontal), let $t(\Gamma)$ be the number of intersections of OPT$'$ with Γ and $T = \sum_\Gamma t(\Gamma)$.

Lemma 15 (Arora (1998))**.** *$T \leq 2 \cdot opt'/(\rho h)$.*

Proof. Let $\ell = (x_1, y_1) \to (x_2, y_2)$ be any leg of OPT$'$. Let $\Delta x = |x_1 - x_2|$ and $\Delta y = |y_1 - y_2|$. The contribution of ℓ to opt$'$ is its length, i.e. $L_\ell = \sqrt{(\Delta x)^2 + (\Delta y)^2}$. Note that due to the scaling, we have $L_\ell \geq 4$. Since the dissecting lines are ρh apart, there are at most $(\Delta x + \Delta y + 2)/(\rho h)$ dissecting lines that intersect with Γ; so the contribution of Γ to T is at most the same amount. Using the Cauchy-Schwarz inequality, we have $2((\Delta x)^2 + (\Delta y)^2) \geq (\Delta x + \Delta y)^2$, which implies

$$(\Delta x + \Delta y + 2)/(\rho h) \leq (\sqrt{2((\Delta x)^2 + (\Delta y)^2)} + 2)/(\rho h) = (\sqrt{2} \cdot L_\ell + 2)/(\rho h).$$

It suffices to show $\sqrt{2} \cdot L_\ell + 2 \leq 2L_\ell$; this is easily seen to be true because $L_\ell \geq 4$. Therefore, if we add these inequalities for all legs ℓ of OPT$'$, we get the lemma's statement as the result. □

The following lemma is essentially the same as the one in the case of point TSP (except we have different stopping points):

Lemma 16 (Arora (1998))**.** *Considering the randomness of the dissecting lines, with probability of at least $\frac{1}{2}$, there exists a portal-respecting solution for $\mathcal{I}'$ with cost at most $(1+\varepsilon) \cdot opt'$ for portal parameter $m = O(\frac{1}{\varepsilon} \cdot \log \frac{N}{\rho h})$*

Proof. The proof is similar to the survey in Vazirani (2003). Consider any dissecting line Γ of level j and focus on the intersections of OPT$'$ with that line. Consider any leg $\ell = ab$ of OPT$'$ which intersects Γ, say, at a point q and suppose p is the nearest portal of Γ to q. Replace ℓ with with two new "legs" $\ell_1 = ap$ and $\ell_2 = pb$. Let d be the distance of q to p. Using triangle inequality, it can be seen that $\ell_1 + \ell_2 \leq \ell + 2d$; meaning the additional cost for going through portal p is at most $2d$. The distances between the portals on level j line Γ are $d_j = \frac{N}{2^j m}$, and clearly $d \leq d_j$. Recall that OPT$'$ intersects with Γ, $t(\Gamma)$ times. Thus, the expected increase in cost for any dissecting line Γ is at most

$$\begin{aligned}
\sum_{j=1}^{\log N/\rho h} \Pr[\Gamma \text{ is level } j] \cdot t(\Gamma) \cdot 2 \cdot \frac{N}{2^j m} &\leq \sum_{j=1}^{\log N/\rho h} \frac{2^{j-1}}{N/\rho h} \cdot t(\Gamma) \cdot 2 \cdot \frac{N}{2^j m} \\
&= \frac{\rho h}{m} \cdot \sum_{i=1}^{\log N/\rho h} t(\Gamma) \\
&= \frac{\rho h}{m} \cdot \log \tfrac{N}{\rho h} t(\Gamma).
\end{aligned}$$

For $m \geq \frac{4}{\varepsilon} \log \frac{N}{\rho h}$, the last value above is at most $\frac{\varepsilon \rho h}{4} \cdot t(\Gamma)$. Adding all these inequalities over different Γ's gives us $\frac{\varepsilon \rho h}{4} \cdot T$, which according to Lemma 15 is at most $\frac{\varepsilon}{2} \cdot$ opt$'$. Using Markov's inequality the statement of the lemma follows. □

The patching Lemma (stated below) for classic Euclidean TSP holds in our setting as well.

Lemma 17 (The patching Lemma Arora (1998))**.** *For any dissecting line segment τ with length L_τ, if a tour crosses τ more than twice, it can be altered to still contain the original tour, but intersect with τ at most twice with an additional cost not greater than $6L_\tau$.*

Proof. The same proof as in Arora (1998) applies here. □

Observation 4. *A single point can be seen as a 0-length segment. By using Lemma 17, we get that at no additional cost (i.e. extra cost of 6×0), each portal is visited at most twice.*

The next lemma shows the existence of a near-optimum solution that is r-light and portal respecting for $r = O(1/\varepsilon)$:

Lemma 18. *Given the randomness in picking the dissecting lines, with probability at least $\frac{1}{2}$, there is an r-light portal respecting tour for $\mathcal{I}'$ with cost at most $(1+\varepsilon) \cdot opt'$ for $r = O(\frac{1}{\varepsilon})$.*

Proof. This is implied by the *Structure Theorem* in Arora (1998), and the similar proof works here. □

Outer DP Table and Time Complexity. The outer DP is similar to the DP for classic Euclidean TSP except that we need to take care of required portals that are going to be guessed and passed down to the subproblems. Note that there are $O(n)$ subproblems in each level of the dissection tree, and so a total of $O(n \log n)$ squares to consider. For each square S with $4m$ portals around it, we guess a subset of portals on the horizontal sides of S to be required. The number of such guesses is 2^{2m} where $m = O(\frac{1}{\varepsilon} \cdot \log \frac{N}{\rho h}) = O(\log n/\varepsilon)$. There are $(4m+1)^{4r}$ guesses for up to $4r$ portals to be chosen for an r-light portal respecting, and at most $(4r)!$ for the pairings of these portals. So the size of the DP table is at most $O(n \log n \cdot 2^{2m} \cdot (4m+1)^{4r} \cdot (4r)!) = O(n \log^{O(r)} n)$.

The DP table is filled bottom up. The base cases are when we have a square of side length $\rho \cdot h$. These subproblems are solved using the inner DP described in the next section. For every other square S that is broken into 4 squares $S_1, \ldots, S_4$, we solve the subproblem of S after we have solved all subproblems for $S_1, \ldots, S_4$. The way we combine the solutions from those of the sub-squares to obtain the solution for S is very much like the classic point TSP. However, we have to extend the solutions so that the line segments that were intersecting the horizontal dissecting line that split S, are now fully covered by the guessed required portals for $S_1, \ldots, S_4$. More specifically, suppose Γ is the horizontal dissecting line that corresponds to a cover-line C_τ from group G_{j^*} (and hence we removed all the segments crossing C_τ and instead made some of the portals along C_τ as required). We add those segments of the instance back, and we extend the solutions from the require portals to travel left and right to cover these segments. Similar to the classic TSP, the total time to fill in the outer DP table is $O(n \log^{O(r)} n)$.

3.2.2. *Inner DP.* Recall that each base case of the quad-tree decomposition is a subproblem defined on a square S with size $\rho h \times \rho h$, and has $4m$ portals around it. Since we assume the solution we are looking for is r-light, it means the instance defined by S has also a set P of size at most $4r$ of portal pairs (where $r = O(1/\varepsilon)$). Each pair $(p_i, q_i) \in P$ specifies that the solution restricted to S, has a p_i, q_i-path. We are also given a guessed subset Q of the portals around S (specifically on the top and bottom side of S) as the required portals. The goal is to find a minimum cost collection of paths that start/end at the given set of portal pairs P that cover all the line segments in S, as well as visit all the required portals in Q. Let us denote this instance by (S, P, Q). Note that by Theorem 2, Lemma 14, and Lemma 18, there is a near-optimum solution such that it is r-light for each square of the dissection, is portal respecting, covers all the required portals, and has shadow bounded by $O(1/\varepsilon^2)$. Also using Lemma 10, the length of any pure reflection sequence in it is bounded by $O(1/\varepsilon)$. We describe the inner DP to find an optimum solution with bounded shadow (and pure reflection sequence bounded to $O(1/\varepsilon)$ elements) restricted to subproblem (S, P, Q). For square S, let us use OPT_S and opt_S to denote such a bounded shadow optimum solution and its value, respectively.

Informally, the DP is a (nontrivial) generalization of the DP for the classic (and textbook example) bitonic TSP in which the shadow is 2. In our case, the shadow is $O(1/\varepsilon^2)$. We are going to consider a sweeping vertical line Γ in S (that moves left to right) and "guess" the intersections of OPT_S with it.

We define an *event point set* in the following way:

Definition 11 (Event Point)**.** *Given a subproblem triplet (S, P, Q), each line segment in S is in the event point set. Also, each portal that is on a horizontal side of S and is either in Q, or participates in a pair of P, is also in the event point set.*

We consider an ordering of all the elements in the event point set from left to right (i.e. increasing x-coordinate), say $v_1, v_2, \ldots, v_{n_S}$, where n_S is the number of event points; note that $n_S = O(n)$. There are $n_S - 1$ equivalent classes for positions of Γ, where each class corresponds to when Γ is located between v_i, v_{i+1}. A sweep line between v_i, v_{i+1} is denoted by Γ_i. Since the shadow of OPT_S is bounded, the intersection of Γ_i with OPT_S has a low complexity. We will give a more concrete explanation of that complexity below.

Recall Observation 1 and the types of points in a solution (straight point, break point, or reflection point). Also recall the definition of a pure reflection point (a reflection point that is not at a tip of a segment of the instance). Consider the global optimum solution that is r-light and portal respecting with bounded shadow and bounded pure reflection sequence that also covers the required portals of each square. Suppose $p_{a_1}, p_{a_2}, \ldots, p_{a_k}$ is the sequence of points in S visited by OPT_S in this order that are *not* a straight point nor a pure reflection point; so each of them is a break point (tip of a segment) or perhaps a required portal

in Q, or a portal in P (i.e. is an entry or exit point in some pair belonging to P). So any point visited by OPT_S between $p_{a_i}, p_{a_{i+1}}$ (if there is any) is either a straight point or a pure reflection point. We define subpaths of OPT_S named *large legs* as follows:

Definition 12 (Large Leg)**.** *The path of OPT_S from p_{a_i} to $p_{a_{i+1}}$ is a large leg. Each large leg starts and ends from a portal or a tip of a segment, and all the points in between are either straight points or pure reflection points.*

It follows from Lemma 10 that the number of pure reflection points in each large leg is bounded by $O(1/\varepsilon)$. Each large leg can be guessed by making at most $O(1/\varepsilon)$ guesses for segments or points: guess the two end-points of the large leg (which are either portals or tips of segments), then guess at most $O(1/\varepsilon)$ segments that have pure reflection points on them; once we guess the two end-points and the segments for pure reflections, the pure reflection points are uniquely determined. Since there are $O(n^2)$ choices for the end-points and $O(n^{1/\varepsilon})$ choices for the segments of pure reflection points, the total number of possible large legs is bounded by $n^{O(1/\varepsilon)}$. Now since we assume OPT_S has bounded shadow of $O(1/\varepsilon^2)$, for any sweep line Γ_i, there are at most $O(1/\varepsilon^2)$ large legs of OPT_S that can cross Γ_i.

So for a fixed i (and sweep the line Γ_i), let $\mathcal{L}_i = L_1, \ldots, L_\sigma$ be the sequence of large legs ($\sigma = O(1/\varepsilon^2)$) of OPT_S that cross Γ_i; where each large leg is specified by the end-points as well as the intermediate segments for pure reflections (if there are any). Then the number of possible choices for $\mathcal{L}_i$ is $n^{O(1/\varepsilon^3)}$. Given i and $\mathcal{L}_i$, let S_i^L, S_i^R be the left and right part of S (cut by Γ_i). If we ignore the segments covered by $\mathcal{L}_i$ in S_i^L, and consider the end-points of each L_j as portals too, then the restriction of OPT_S to S_i^L is a collection of paths that start/end at portals of P in S_i^L or end-points of L_j's in S_i^L that cover all the segments in S_i^L not already covered by $\mathcal{L}_i$, as well as points in $Q \cap S_i^L$. More specifically, each part of OPT_S in S_i^L is a path that starts at a p_j for a pair $(p_j, q_j) \in P$, or at an end-point of L_k that is in S_i^L and ends at a point $p_{j'}$ (or $q_{j'}$) of another pair in P that is also in S_i^L, or at another end-point of some $L_{k'}$ that is in S_i^L. So this induces some pairs of points, denoted by P_i^L:

Definition 13 (Path-wise Pairing P_i^L)**.** *Set P_i^L of pairs of points is said to be the **path-wise pairing** for S_i^L, if there is a path in S_i^L between the two points of any given pair $(a, b) \in P_i^L$. Furthermore, each point in a pair $(a, b) \in P_i^L$ is either a portal in S_i^L that is part of a pair in P, or is an end-point of a large leg L_j that is in S_i^L.*

For any such point in S_i^L, say p, there must be a pair in P_i^L containing that point. We also assume $(p, p) \in P_i^L$, and if p is an end point of a large leg in $\mathcal{L}_i$ or S_i^L, and if q is the other end point of that large leg, then $(p, q) \in P_i^L$.

We say a set of pairs P_i^L is not *promising* if given $\mathcal{L}_i$, there is no feasible solution in the entire S whose restriction to S_i^L defines subpaths consistent with P_i^L (i.e. they start and end on the same pairs as specified by P_i^L). Otherwise, we consider it promising. For example if $(p_j, q_j) \in P$, both p_j, q_j belong to S_i^L, and if $(p_j, u), (q_j, v) \in P_i^L$ where u is one end of a long leg L_1 and v is one end of a long leg L_2, it must be the case that it is possible to have a path from the other end of L_1 to the other end of L_2. This would be impossible if, for instance, those other ends of L_1, L_2 are paired up with other portals in P_i^L. Note that since there are at most $4r$ pairs in P and $O(1/\varepsilon^2)$ end-points in $\mathcal{L}_i$, the number of possible choices for P_i^L is $(1/\varepsilon)^{O(1/\varepsilon)}$. Also, a given P_i^L (together with $\mathcal{L}_i$), it can be checked if P_i^L is promising or not in poly-time in n.

This suggests how we can break the instance (S, P, Q) into polynomially many sub-instances. For a fixed i, guess $\mathcal{L}_i$ among all those with shadow $O(1/\varepsilon^2)$, break S into S_i^L, S_i^R, let $Q_i^L = Q \cap S_i^L$, and guess the new pairs P_i^L (for S_i^L) that are promising. We solve (S_i^L, P_i^L, Q_i^L) for each S_i^L, P_i^L, Q_i^L obtained this way. We can solve each such subproblem assuming we have solved all subproblems defined by each Γ_j for $j < i$. So formally, let us define a *configuration*:

Definition 14 (Configuration)**.** *A **configuration** is a vector $(i, \mathcal{L}_i, P_i^L)$ where the components are:*

- *i (indicating Γ_i and defining S_i^L),*
- *The large legs of OPT_S crossing Γ_i, denoted by $\mathcal{L}_i$, $|\mathcal{L}_i| = O(1/\varepsilon^2)$,*
- *The pairing P_i^L defined by $\mathcal{L}_i$, P, and the restriction of OPT_S to S_i^L.*

This configuration (see Figure 11), defines a subproblem: Suppose $\mathcal{L}_i$ is a given set of large legs crossing Γ_i. Find a collection of paths in S_i^L such that P_i^L specifies the start/end of these paths (and is promising), such that these paths cover all the segments in S_i^L (excluding those already covered by $\mathcal{L}_i$), and also cover all the points in $Q \cap S_i^L$, with shadow at most $O(1/\varepsilon^2)$.

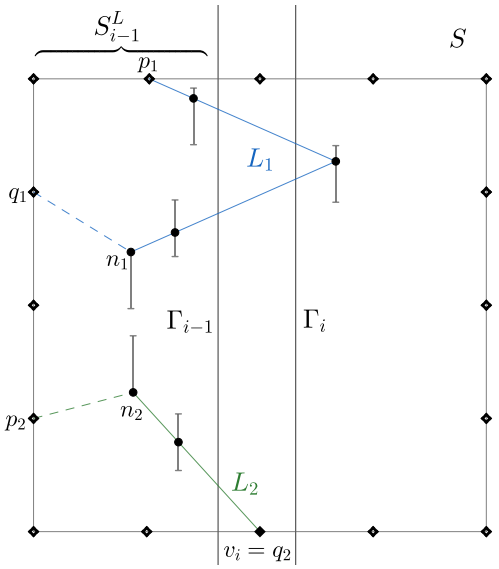


FIGURE 11. An example of an event point v_i and vertical lines Γ_{i-1}, Γ_i from two consecutive equivalent classes in square S. In this figure, $\mathcal{L}_{i-1} = L_1, L_2$ and $\mathcal{L}_i = L_1$; plus, it is the case that $(p_2, n_2) \in P^L_{i-1}, (q_1, n_1) \in P^L_{i-1}$, and $(q_1, n_1) \in P^L_i$

The cost of this solution is defined to be the sum of the costs of all the edges that are entirely (i.e. both end-points) in S^L_i (including those legs of a large leg in $\mathcal{L}_i$ that are entirely in S^L_i, but not those that are crossing Γ_i). Entry $A[i, \mathcal{L}_i, P^L_i]$ of the inner DP, stores the minimum cost of such a solution. Recall that there are $n_s = O(n)$ choices for i (and so for Γ_i), $n^{O(1/\varepsilon^3)}$ choices for $\mathcal{L}_i$, and $(1/\varepsilon)^{O(1/\varepsilon)}$ choices for P^L_i. So there are $n^{O(1/\varepsilon^3)}$ possible configurations, which is the size of our DP table as well.

We fill in the entries of this table $A[.,.,.]$ for increasing values of i. For $i = O(1)$, $A[i,.,.]$ can be computed exhaustively in $O(1)$ time.

For any other value of i, we compute $A[i, \mathcal{L}_i, P^L_i]$ by considering various subproblems $(i-1, \mathcal{L}_{i-1}, P^L_{i-1})$ that are *consistent* (formally explained soon) with $(i, \mathcal{L}_i, P^L_i)$. Consider event point v_{i-1}; it is either a segment or a portal that is between Γ_{i-1} and Γ_i; which means it does not belong to S^L_{i-1}, but belongs to S^L_i. Consider the solution for $(i, \mathcal{L}_i, P^L_i)$, and the legs (in that solution) that visit v_i. In case v_i is a start/end portal in P, there is one leg incident to v_i; if $v_i \in Q$ there are two legs incident to v_i, and if v_i is a segment, there are two legs that are incident to a point v'_i on that segment. If there is one leg only (v_i is a start/end portal), call that leg ℓ_i, and if there are two legs, call them ℓ_{i-1}, ℓ_i. Depending on whether these legs cross Γ_{i-1} or Γ_i, we have the following situations, which are the *consistent* outcomes:

(1) v_i **is a start/end portal**, we consider 2 different subcases:
   (a) ℓ_i **crosses** Γ_{i-1} **but not** Γ_i: Say $\ell_i = v_i u$, where u is a point in S^L_{i-1}. In this case, there is a large leg $L \in \mathcal{L}_{i-1}$ with one end-point v_i. Then if L crosses Γ_i, it means L is a large leg in $\mathcal{L}_i$. If L does not cross Γ_i, then $\mathcal{L}_i = \mathcal{L}_{i-1} \setminus L$. We consider both possibilities and in each case, consider P^L_{i-1}'s that are consistent with P^L_i and set $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\} + ||\ell_i||$.
   (b) ℓ_i **crosses** Γ_i **but not** Γ_{i-1}**:** In this case, there is a large leg $L \in \mathcal{L}_i$ that starts with ℓ_i and does not cross Γ_{i-1}, so does not belong to $\mathcal{L}_{i-1}$. All the other large legs in $\mathcal{L}_{i-1}$ and $\mathcal{L}_i$ are the same (as there is no other event point between Γ_{i-1} and Γ_i), and P^L_i and P^L_{i-1} are consistent. Then $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\}$.

(2) $v_i \in Q$, we consider 3 different subcases:
   (a) ℓ_{i-1}, ℓ_i **both cross** Γ_{i-1} **but not** Γ_i**:** In this case, there are two large legs $L, L' \in \mathcal{L}_{i-1}$ that both end at v_i, say L contains ℓ_i and L' contains ℓ_{i-1}. If L crosses Γ_i, then L is a large leg in $\mathcal{L}_i$ as well, similarly for L'. The other large legs of $\mathcal{L}_i$ and $\mathcal{L}_{i-1}$ are the same, and P^L_{i-1} is consistent with P^L_i. We set $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\} + ||\ell_i|| + ||\ell_{i-1}||$.

(b) ℓ_{i-1}, ℓ_i **both cross** Γ_i **but not** Γ_{i-1}**:** This similar to the previous case. There are two legs $L, L' \in \mathcal{L}_i$ that both start at v_i, say L contains ℓ_i and L' contains ℓ_{i-1}. If L crosses Γ_{i-1}, then L is a large leg in $\mathcal{L}_{i-1}$ as well, similarly for L'. The other large legs of $\mathcal{L}_i$ and $\mathcal{L}_{i-1}$ are the same and P^L_{i-1} is consistent with P^L_i. In this case, $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\}$.

(c) **Exactly one of** ℓ_{i-1}, ℓ_i **crosses** Γ_{i-1} **and one crosses** Γ_i**:** Say ℓ_{i-1} crosses Γ_{i-1}, and ℓ_i crosses Γ_i. So ℓ_{i-1} will be the last leg of a large leg $L \in \mathcal{L}_{i-1}$, and ℓ_i will be the first leg of a large leg $L' \in \mathcal{L}_i$. If L does not cross Γ_i, then L is not in $\mathcal{L}_i$ at all. Similarly, if L' doesn't cross Γ_{i-1}, then L' isn't a large leg in $\mathcal{L}_{i-1}$. We consider both possiblities (i.e. consider sets $\mathcal{L}_{i-1}$ that are consistent with one of these cases). $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\} + ||\ell_{i-1}||$.

(3) v_i **is a segment**: Subcases are similar to the previous case; let v'_i be the intersection point of OPT_S with v_i:

(a) ℓ_{i-1}, ℓ_i **both cross** Γ_{i-1} **but not** Γ_i**:** If v'_i is a tip, then ℓ_{i-1} is the last leg of a large leg $L \in \mathcal{L}_{i-1}$, and ℓ_i is the last leg of another large leg $L' \in \mathcal{L}_{i-1}$. Depending on whether L (L') crosses Γ_i, it can be a large leg in $\mathcal{L}_i$ or not. We consider both possibilities. If v'_i is not a tip, then it must be a pure reflection, so there must be a large leg $L \in \mathcal{L}_{i-1}$ that contains this as a pure reflection. That large leg may or may not belong to $\mathcal{L}_i$. We consider all these possibilities (i.e. those $\mathcal{L}_{i-1}$ consistent with these), and also for each case consider a P^L_{i-1} consistent with P^L_i. Then set $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\} + ||\ell_{i-1}|| + ||\ell_i||$.

(b) ℓ_{i-1}, ℓ_i **both cross** Γ_i **but not** Γ_{i-1}**:** If v'_i is a tip, then ℓ_{i-1} is the first leg of a large leg $L \in \mathcal{L}_i$, and ℓ_i is the first leg of another large leg $L' \in \mathcal{L}_i$. Depending on whether L (L') crosses Γ_{i-1}, it can be a large leg in $\mathcal{L}_{i-1}$ or not. We consider both possibilities. If v'_i is not a tip, then it must be a pure reflection, so there must be a large leg $L \in \mathcal{L}_i$ that contains this as a pure reflection. That large leg may or may not belong to $\mathcal{L}_{i-1}$ depending on whether it crosses Γ_{i-1} or not. We consider all these possibilities, and also for each case consider a P^L_{i-1} consistent with P^L_i. Then set $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\}$.

(c) **Exactly one of** ℓ_{i-1}, ℓ_i **crosses** Γ_{i-1} **and one crosses** Γ_i**:** In this case, v'_i must be a tip or a straight point. Say ℓ_{i-1} crosses Γ_{i-1}, and ℓ_i crosses Γ_i. If v'_i is a tip, then ℓ_{i-1} is the last leg of a large leg $L \in \mathcal{L}_{i-1}$, and ℓ_i is the first leg of a large leg $L' \in \mathcal{L}_i$. L may cross Γ_i (in which case it also belongs to $\mathcal{L}_i$), also L may cross Γ_{i-1} in which case belongs to $\mathcal{L}_{i-1}$. We consider these possibilities. If v'_i is a straight point, then both ℓ_{i-1}, ℓ_i are part of a large leg $L \in \mathcal{L}_{i-1}$, and L belongs to $\mathcal{L}_i$ as well. We consider all these cases and consistent P^L_{i-1}, P^L_i and set $A[i, \mathcal{L}_i, P^L_i] = \min_{P^L_{i-1}, \mathcal{L}_{i-1}} \{A[i-1, \mathcal{L}_{i-1}, P^L_{i-1}]\} + ||\ell_{i-1}||$.

Consistent Subproblems. The consistency of a subproblem by configuration $(i, \mathcal{L}_i, P^L_i)$, with a previous subproblem by configuration $(i-1, \mathcal{L}_{i-1}, P^L_{i-1})$, comes down to one of the cases mentioned in the previous section. In each subcase, we only need to define what we mean by consistent between P^L_i and P^L_{i-1}.

We say P^L_i as a part of the configuration $(i, \mathcal{L}_i, P^L_i)$, and P^L_{i-1} as a part of the configuration $(i-1, \mathcal{L}_{i-1}, P^L_{i-1})$ are consistent if for any pair $(a, b) \in P^L_i$:

- If both a, b are in S^L_{i-1}, then either:
  - $(a, b) \in P^L_{i-1}$, or
  - (When $v_i \in Q$ or when v_i is a segment containing a pure reflection) There is a large leg $L_j \in \mathcal{L}_{i-1} \cup \mathcal{L}_i$ with end points p_1, p_2 corresponding to (i.e. having an intersection with) the event point v_i, such that $(a, p_1), (b, p_2) \in P^L_{i-1}$, or
  - (When $v_i \in P$ or v_i is a segment containing a non-pure reflection or a break point) There are two large legs (in $\mathcal{L}_{i-1} \cup \mathcal{L}_i$) that have v_i as an end point, and have another end point, say respectively p_1 and p_2, such that $(a, p_1), (b, p_2) \in P^L_{i-1}$.
- If both a, b are not in S^L_{i-1}, then it means that either a or b, say a, corresponds to the event point v_i. This means either a is a portal ($\in P \cup Q$) between Γ_{i-1} and Γ_i, or a is a tip of the segment corresponding to v_i. In either case, there is at least a large leg $L_j \in \mathcal{L}_{i-1} \cup \mathcal{L}_i$ that has a as one of its end points. There can be at most two such large legs; say p_1 and possibly p_2 are the other ends of these at most two large legs. Then it must be the case that (either) $(b, p_1) \in P^L_{i-1}$ (or $(b, p_2) \in P^L_{i-1}$).

3.3. **Wrap-up: Proof of Theorem 1.** We finalize the proof of our algorithm for the case of unit length segments, and then extend the proof to the case of similar-length segments.

Theorem 4. *There is a $(1+\varepsilon)$-approximation algorithm for TSPN over n vertical unit-length line segments that runs in time $n^{O(1/\varepsilon^3)}$.*

Proof. Take any instance of the problem. As described at the beginning of this section, we first scale the instance (at a loss of $(1+\varepsilon)$) so that all segments have integer coordinates. We employ the hierarchical decomposition of Arora using dissecting lines as described in Subsection 3.1, and drop the line segments crossing horizontal dissecting lines as described in Subsection 3.1.1. We require a subset of portals around each square S of the dissectioning to be covered in the subproblems as described in the outer DP in Subsection 3.2.1. Lemma 14 shows that we lose at most another $(1+\varepsilon)$ factor in doing so. At the leaf level of our decomposition, we need to solve instances where each square has sides of length $\rho \cdot h$. Note that as discussed in the first paragraph of Subsection 3.2.2, for any base square of the dissection, using Theorem 2, Lemma 14, Lemma 18, and Lemma 10, there is a near-optimum solution such that it is portal respecting, r-light for $r = O(\frac{1}{\varepsilon})$, covers all the required portals, has a shadow bounded by $O(1/\varepsilon^2)$, and the length of any pure reflection sequence in it is bounded by $O(1/\varepsilon)$. The inner DP describes how to find such a solution. Note that the size of the inner DP table is $n^{O(1/\varepsilon^3)}$. To compute each entry, we may consider (at worst) all other entries, and so the time complexity of computing the table for each square S is at most $n^{O(1/\varepsilon^3)}$. Given that the number of squares at the leaf nodes of the decomposition is $O(n \log^{O(r)} n)$, the total time for the inner and outer DP is $n^{O(1/\varepsilon^3)}$. □

We discuss how the result presented for unit-length line segments in Theorem 2 can be extended to the case that line segments have length ratio $\lambda = O(1)$, and obtain a PTAS for it. In the case of segments with lengths in $[1, \lambda]$, for every strip of height 1, we still have some top and bottom segments and we might have some line segments that completely span the height of the strip. Let's call these segments *full segments* of a strip. We claim that whenever we change the solution in the proof of Theorem 2 to one that has a bounded shadow, the full segments of the strip remain covered. These changes are done in Lemmas 10 and 9. For each of these cases, any new subpath (with smaller shadow) that replaces a subpath of larger shadow, will travel the same interval in the x-coordinate, and hence any full segment covered by the original path, remains covered by the new path.

Next, when we scale the instance, we get line segments with length between $[\rho, \lambda\rho]$. Now we do our hierarchical decomposition until base squares have side length of $\lambda\rho h$, so the space between two cover-lines in the same group is $\lambda\rho h$ instead of ρh. Lemma 12 holds with bound opt $\geq \frac{\rho \cdot |\mathcal{B}|}{6\lambda}$. This implies Lemma 13 holds if j is chosen from $[1 \dots h\lambda]$. It is straight-forward to check that Lemma 14 holds with the same ratio. For the inner DP, noting that the instance we start from has height $\rho\lambda h$, the shadow is bounded by $O(\lambda/\varepsilon^2)$. The same DP works but the runtime will be $n^{O(\lambda/\varepsilon^3)}$. This implies we get a PTAS with the same run time which completes the proof of Theorem 1. □

4. Proofs of the Major Lemmas

In this section, we provide the proofs of the four major lemmas we had, namely Lemmas 8, 9, 10, and 11; as well as the proof of Theorem 3.

4.1. Properties of an Optimum Solution.

The goal of this subsection is to prove Lemma 8. Before that, we need to state some further lemmas and definitions. We combine these in Subsection 4.1.1 to build a complete proof.

Definition 15. *Let OPT_τ be the restriction of OPT to any strip S_τ. We say a segment $s \in S_\tau$ is **exclusively covered** by some path $P \in OPT_\tau$ if P covers s but no other subpath of OPT_τ intersects with s, i.e. OPT_τ / P doesn't intersect with it.*

Lemma 19. *Each loop with entry points on $C_{\tau+1}$ in OPT_τ (i.e. bottom cover-line of S_τ) must exclusively cover a top segment, or else it must be a cover-line loop. Analogous argument holds for loops that have entry points on C_τ.*

Proof. Suppose P is a loop with entry points e_1, o_1 on $C_{\tau+1}$ that does not exclusively cover a point on a top segment. This implies if we change it to cover only bottom segments in S_τ, then the solution remains feasible. Let s_ℓ and s_r be the left-most and right-most bottom segments that P covers, let q_ℓ, q_r be intersections of s_ℓ and s_r with $C_{\tau+1}$, respectively. Replace P with e_1, q_ℓ, q_r, o_1 and then short-cut e_1, o_1 like the way we

argued for cover-line loops after Definition 7. So we obtain a path that is shorter than the original, but is a cover-line loop and covers all the (bottom) segments P was covering. □

Corollary 3. *If P is a non-cover-line loop with entry points on the bottom cover-line of some strip S_τ, then P has to exclusively cover some top segment in S_τ. Similar argument holds for bottom segments and non-cover-line loops with entry points on the top cover-line.*

Lemma 20. *Suppose that OPT_τ is crossing a vertical line Γ at least two times. Let p_1, p_2 be two such crossings and, L_1 be a subpath of OPT_τ from p_1 to p_2 with no other crossings with Γ. Then there cannot be any other crossings of OPT_τ with Γ on the section p_1p_2 of Γ.*

Proof. Without loss of generality, since L_1 doesn't intersect with Γ other than at points p_1 and p_2, assume that L_1 is on the left of Γ. By way of contradiction, suppose q_1 is another crossing of OPT_τ with Γ such that $y(p_1) < y(q_1) < y(p_2)$. This implies that there is a subpath of OPT_τ inside the region $A = L_1 \cup p_1p_2$ with one end-point being q_1. So there must be another crossing of OPT_τ with the region $A = L_1 \cup p_1p_2$; and since OPT_τ is not self-crossing, that other crossing point with A must be on p_1p_2, call it q_2. Let us denote the subpath of OPT_τ inside A with end-points q_1, q_2 by L_2. Let r_1 be the left-most point on L_1. Since L_1 is a path from a point on Γ to the left of Γ and back to a point on Γ, using Lemma 1, r_1 must be a right reflection point. Similarly, if r_2 is the left-most point on L_2 then r_2 must be a right reflection point, say on segment s_{r_2} (see Figure 12). But since r_2 is inside A, then regardless of whether s_{r_2} is a top segment or a bottom segment it will intersect with L_1, contradicting Lemma 7. □

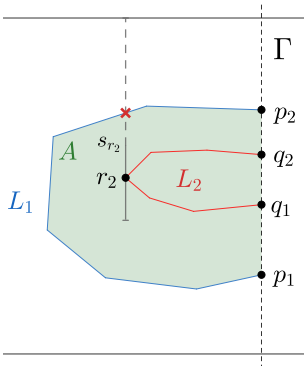


FIGURE 12. Configuration for Lemma 20

The following lemma is a special case of Lemma 20, but since it is used frequently, we state it as a separate lemma.

Lemma 21. *Consider a strip S_τ and OPT_τ (the restriction of OPT within this strip). Let s be any segment in this strip which has a reflection point p_j on it. Without loss of generality, assume s is a top segment and p_j is a left reflection point. Let ℓ_u and ℓ_l be the upper and lower legs of OPT_τ incident with p_j. Then the subpath of OPT_τ starting at p_j and travelling on ℓ_u, will not reach to the right side of s.*

Proof. Suppose the subpath of OPT_τ starting at p_j and travelling along ℓ_u, call it P_u, reaches the right side of s while entirely within strip S_τ. So P_u crosses the vertical line $x = x(s)$ at a point p inside S_τ (different from p_j). This path will be L_1 in the setting of Lemma 20 and p_j, p will be p_1, p_2 of the lemma. Consider the subpath P_l of OPT_τ starting at p_j and following ℓ_l. This subpath is in the region defined by P_u and the vertical line at $x = x(s)$. Since OPT is non-self-crossing, P_l has to exit this area between the lower tip of s and point p. But this will violate Lemma 20. This contradiction results in the statement of the lemma. □

Lemma 22. *Suppose P_1 and P_2 are two ladders of OPT_τ in S_τ with entry points e_1 and e_2 on the bottom cover-line and entry points o_1, o_2 on the top cover-line, respectively, such that $x(e_1) < x(e_2)$, $x(o_1) < x(o_2)$ and both intersect a vertical line Γ to the right of e_1, e_2. Then P_1 is above P_2 to the left of Γ. (symmetric arguments apply to the top cover-line, as well as entry points to the right of Γ)*

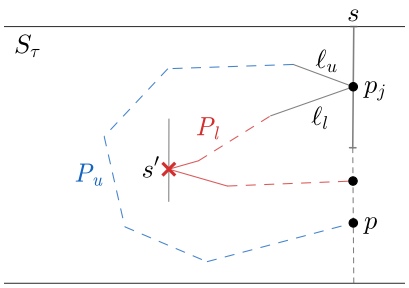


FIGURE 13. In a strip S_τ, the path from the upper leg of a left reflection on a top segment, can't reach to the right of that segment

Proof. By way of contradiction, suppose P_1 is not above P_2 on the left of Γ, so there is a vertical line Γ' to the left of Γ whose top-most intersection is with P_2, say point p on Γ'. Consider the (vertical) segment of Γ' from p to the top cover-line, call it Γ'' and let the subpath of P_2 from e_2 to p be called P_2'. If we cut the strip S_τ along $P_2' \cup \Gamma''$, then e_1 is on one side, and o_1 on the other, which implies L_1 must be crossing $P_2' \cup \Gamma''$, which would be a contradiction (as it would have an intersection point on Γ' higher than p or has to cross P_2'). □

Lemma 23. *Let P be any ladder or loop of OPT_τ in strip S_τ. Let r_{i_1} (on segment s_{m_1}) and r_{i_2} (on segment s_{m_2}) and r_{i_3} (on segment s_{m_3}) be any three consecutive reflections in the orientation of OPT_τ in that order. If $x(r_{i_2}) < x(r_{i_1}) < x(r_{i_3})$ and r_{i_2} is an ascending reflection, then s_{m_1} is a bottom segment and r_{i_1} is an ascending reflection. Symmetric argument applies for r_{i_2} being a descending reflection (for which case s_{m_1} will be a top segment and r_{i_1} will be descending).*

Proof. See Figure 14. According to Lemma 6, since r_{i_1} and r_{i_2} are consecutive reflections with $x(r_{i_2}) < x(r_{i_1})$, then r_{i_1} is a left reflection and r_{i_2} is a right reflection. Let $P_{1,2}$ be the subpath of P from r_{i_1} to r_{i_2}, and $P_{2,3}$ be the subpath of P from r_{i_2} to r_{i_3}. Since r_{i_2} is an ascending reflection, then $P_{1,2}$ contains the lower leg of r_{i_2}, and $P_{2,3}$ contains the upper leg of r_{i_2}.
Since r_{i_1} and r_{i_2} are two consecutive reflections with $x(r_{i_1}) > x(r_{i_2})$, this means that $P_{1,2}$ cannot reach to the left of r_{i_2} or to the right of r_{i_1}; because otherwise, due to the difference in the x-coordinates, $P_{1,2}$ would require an additional reflection between r_{i_1} and r_{i_2}, which isn't possible.
This implies that the entirety of $P_{1,2}$, and specifically r_{i_1}, are in the region defined by $x = x(r_{i_2})$, $x = x(r_{i_1})$, and the path $P_{2,3}$. So $P_{1,2}$ is below $P_{2,3}$ in $I = [x(r_{i_2}), x(r_{i_1})]$.

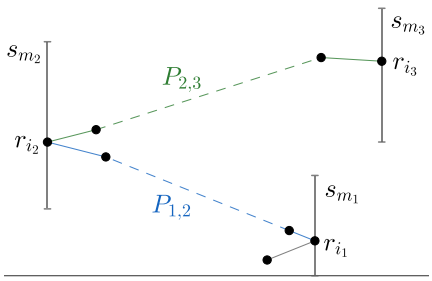


FIGURE 14. Valid arrangement of three consecutive reflections provided the x-coordinate of r_{i_1} is between the x-coordinates of r_{i_2} and r_{i_3}, and r_{i_2} is an ascending reflection. Segments s_{m_2} and s_{m_3} could either be top or bottom segments in this strip, but s_{m_1} must be a bottom segment

Since $x(r_{i_2}) < x(r_{i_3})$ and $P_{2,3}$ is a path between these two reflections, we get that for any $x_0 \in [x(r_{i_2}), x(r_{i_3})]$, there is an intersection between $x = x_0$ and $P_{2,3}$. Now, for the sake of contradiction, assume s_{m_1} is a top segment. Since r_{i_1} is below $P_{2,3}$, this would imply that s_{m_1} is intersecting with $P_{2,3}$. But this is in violation with Lemma 7. Thus, s_{m_1} must be a bottom segment. According to Lemma 21, r_{i_1} cannot be a descending reflection, because otherwise, $P_{1,2}$ would contain the lower leg of r_{i_1}; therefore, the path

$P_{1,2} \cup P_{2,3}$ is a path that contains the lower leg of r_{i_1} and reaches to the right of segment s_{m_1}, which isn't possible. So we conclude that s_{m_1} is a bottom segment and furthermore, r_{i_1} is an ascending reflection. □

Lemma 24. *Suppose P is a loop or ladder of OPT_τ for a strip S_τ and $r_{i_1}, r_{i_2}, r_{i_3}$ are three reflection points visited in this order, but not necessarily consecutively (following orientation of OPT), all are ascending (or all are descending) and are on segments $s_{m_1}, s_{m_2}, s_{m_3}$, respectively. Assume that r_{i_1}, r_{i_3} are left reflections and r_{i_2} is a right reflection and r_{i_2} is to the left of both r_{i_1} and r_{i_3}, i.e. $x(s_{m_2}) < x(s_{m_1})$ and $x(s_{m_2}) < x(s_{m_3})$. Let $P_{0,1}$ be the subpath of P up to r_{i_1}, $P_{1,2}$ be the subpath of P from r_{i_1} to r_{i_2}, $P_{2,3}$ be the subpath of P from r_{i_2} to r_{i_3}, and $P_{3,4}$ be the subpath of P from r_{i_3} to the end of P. Then we cannot have both $P_{0,1}$ and $P_{3,4}$ reach to the left of $x(s_{m_2})$.*

Proof. Each of $P_{1,2}$ and $P_{2,3}$ include a leg of r_{i_2}; Without loss of generality, assume that the lower leg of r_{i_2} is in $P_{1,2}$, and its upper leg is in $P_{2,3}$ (i.e. assume that r_{i_2} is an ascending reflection). We take two cases based on whether s_{m_2} is a top segment or a bottom segment:

- s_{m_2} **is a top segment:** Path $P_2^u = P_{2,3} \cup P_{3,4}$ includes the upper leg of r_{i_2} (a right reflection) on a top segment s_{m_2}, so we can use the result of Lemma 21 to conclude that P_2^u and particularly $P_{3,4}$ can't reach to the left of s_{m_2}.
- s_{m_2} **is a bottom segment:** Path $P_2^l = P_{0,1} \cup P_{1,2}$ includes the lower leg of r_{i_2} (a right reflection) on a bottom segment s_{m_2}. Again, using Lemma 21, we get the same result that P_2^l and consequently $P_{0,1}$ can't reach to the left of s_{m_2}.

□

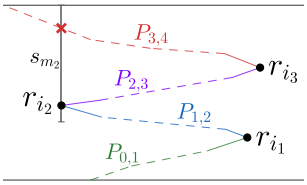


FIGURE 15. Configuration of Lemma 24 when s_{m_2} is a top segment

4.1.1. *Proof of Lemma 8: Vertically Partitioning the Solution in Each Strip.* This subsection is dedicated to the proof of Lemma 8. As mentioned before, we show that the vertical line at which the largest shadow for P happens, can intersect with at most two sinks and a zig-zag. So the shadow of P is within $O(1)$ of the maximum shadow of the zig-zags and sinks along it.

If $q \leq 2$, then the correctness of lemma follows easily; so let's assume $q > 2$. Starting from $i = 1$, find the largest j such that the sequence $r_i, \ldots, r_j$ are all monotone, i.e. are all ascending or all are descending reflections. This will be the first part. We set $i = j + 1$ and, again, find the largest j such that $r_i, \ldots, r_j$ are monotone; this becomes the 2nd part. We repeat this procedure.

So we find a partition into maximal sub-sequences of consecutive reflection points r_a's such that each sub-sequence contains only ascending or only descending reflections; each part might have just one point and the subpath path between the last reflection point of a part and the first reflection point of the next part has shadow 1 (since change in shadow can only happen if there is a reflection point by Lemma 3).

The proof has two parts, we first show that each part can only have up to two sinks and possibly a zig-zag in between them, and then we show that for any vertical line, there is at most one part intersecting with it. Since the subpath from one part to another is a path between two consecutive reflections (and has shadow 1) the statement of the lemma will follow. We will use the following claim throughout this proof:

Claim 1. *For any subpath P of OPT_τ that is either a loop or a ladder, let r_j (on segment s_m) and $r_{j'}$ (on segment $s_{m'}$) be any two consecutive reflections along P. Without loss of generality assume that r_j is an ascending left reflection and in the orientation of OPT_τ, r_j comes before $r_{j'}$. If both s_m and $s_{m'}$ are bottom segments, then the subpath of P up to r_j (which includes the lower leg ℓ_j) can only contain reflection points lying on bottom segments. Analogous statement holds when both $s_m, s_{m'}$ are top segments.*

Proof of claim. According to Lemma 6, $r_{j'}$ is a right reflection and $s_{m'}$ is to the left of s_m. Let P_j be the subpath of P from r_j to $r_{j'}$. Refer to the area of S_τ enclosed by s_m and $s_{m'}$ and below P_j by A_j; then ℓ_j lies inside A_j (see Figure 16). Let the subpath of P ending with leg ℓ_j be called P'_j. So P'_j is entirely within A_j as it cannot intersect with either of $s_m, s_{m'}$ (due to Lemma 7, since they both have reflection points) and P'_j cannot intersect P_j other than at r_j (since the solution is not self-crossing). So P'_j is below P_j within A_j. This implies any top segment that intersects P'_j must also intersect P_j due to Observation 2. So P'_j cannot have a reflection on a top segment by Lemma 7. So P'_j can have reflection points only on bottom segments. □

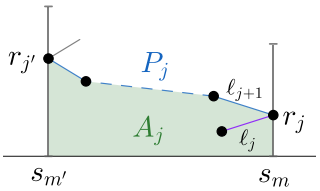


FIGURE 16. (Claim 1) A_j is the area in S_τ surrounded by segments $s_m, s_{m'}$ and the subpath P_j

Now back to the proof of the lemma; we prove the following two parts:

Each partition can have up to two sinks and a zig-zag. Consider any part in the partition we defined before, which is a maximal sub-sequence of consecutive reflections that are all ascending or all descending. Our goal is to show this part is concatenation of a sink (possibly empty), followed by a zig-zag (possibly empty), followed by a sink (possibly empty), where the last point of the first sink is common with the first point of the zig-zag, and the last point of the zig-zag is common with the first point of the last sink. For simplicity, suppose the sequence of this part is $\mathcal{R} = r_1, \ldots, r_k$. Without loss of generality, assume $\mathcal{R}$ contains only ascending reflections and that the first one, r_1, is on a bottom segment. If all r_i's belong to bottom segments, then $\mathcal{R}$ is a sink and we are done. Otherwise, let j be the first index such that r_j is on a top segment (i.e. $r_1, \ldots, r_{j-1}$ are all on bottom segments). If $j = 2$, i.e. r_1 was a bottom and r_2 is a top segment, then the first sink is empty and this part starts with a zig-zag. If $j > 2$, then $r_1, \ldots, r_{j-1}$ is a sink. We argue that starting at $j' = \max\{1, j-1\}$, we can form a zig-zag. Let m be the largest index such that $r_{j'}, r_{j'+1}, \ldots, r_m$ is a zig-zag, i.e. the reflection points alternate between top and bottom segments. If no such m exists, it means $r_{j'}, \ldots, r_k$ all belong to top segments, giving us a sink; so this together with the first possible sink gives us two sinks at most, concluding the lemma. If $m = k$, then the partition has (up to) a single sink followed by a zig-zag, and we're done. Otherwise, $m < k$, meaning $r_{m+1} \in \mathcal{R}$. Since any zig-zag has at least 2 reflections, we have $m \geq 2$, meaning $r_{m-1} \in \mathcal{R}$ and since alternation between top and bottom ends at r_m, it means r_m and r_{m+1} are both either on top segments or both on bottom segments. We show that they can't be both on bottom segments. For the sake of contradiction, assume otherwise, i.e. both r_m and r_{m+1} are on bottom segments (and we assumed they are ascending). Also, we know r_{m-1} is on a top segment (as it must be different from r_m). This violates Claim 1; because r_{m-1} is on a top segment and is on the subpath of OPT_τ reaching r_m. Thus, both r_m, r_{m+1} are on top segments.
Without loss of generality, assume that r_m is a left reflection; Lemma 7 implies r_{m+1} is right reflection with $x(r_{m+1}) < x(r_m)$. Let the path from r_m to r_{m+1} be P_m. Let A_m denote the area (of S_τ) bounded by P_m and between the segments containing r_m and r_{m+1}. If $\ell_{m''}, \ell_{m''+1}$ are the legs incident to r_{m+1} in the orientation of OPT, then $\ell_{m''+1}$ lies inside A_m (see Figure 17). Once again, using Claim 1, we get that there can't be any reflections in the subpath in P starting at r_{m+1} through $\ell_{m''+1}$ that lie on a bottom segment. This implies all of $r_m, r_{m+1}, \ldots, r_k$ lie on top segments. Since all the remaining reflections are on top segments and all are ascending, this by definition means they form a sink. Thus, in total, we have up to a (bottom) sink, a zig-zag, and a (top) sink in this partition, concluding the first part of the proof.

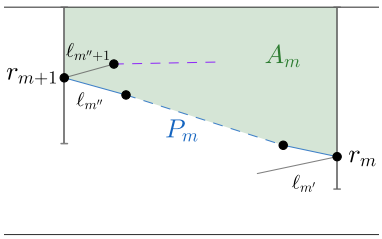


FIGURE 17. The upper leg of r_{m+1} lies inside A_m, and therefore, so does the rest of the path of OPT_τ until r_k

Any vertical line can intersect at most one part. Recall that $r_1, r_2, \ldots, r_q$ denotes the sequence of *all* the reflection points on P (in strip S_τ). Let's call this $\mathcal{R}$. If $\mathcal{R}$ is made of only ascending or only descending reflections, we are done as it will have only one part in the partition. Otherwise, there must be two consecutive reflections $r_i, r_{i+1} \in \mathcal{R}$ such that one is an ascending reflection, but the other is descending. Suppose i is the first index that this happens. So the subpath from r_1 to r_i is one part, and r_{i+1} is the start point of another part. Note that the path from r_i to r_{i+1} has no reflection points; hence has shadow 1 because of Lemma 3. Without loss of generality, assume r_i is a right reflection and is an ascending reflection. Then r_{i+1} is descending and according to Lemma 6, it must be a left reflection as well with $x(r_i) < x(r_{i+1})$. Let P_i be the subpath of P from r_i to r_{i+1}. Since $q > 2$, we either have $i > 1$ or (if $i = 1$ then) $i+1 < q$; meaning $r_{i-1} \in \mathcal{R}$ or $r_{i+2} \in \mathcal{R}$. In other words, there either is a reflection in $\mathcal{R}$ before r_i, or there is a reflection after r_{i+1}. Assume the first case holds, similar argument applies to the second one. Since r_i is a right reflection, using Lemma 6, we get that r_{i-1} is a left reflection with $x(r_i) < x(r_{i-1})$. We claim that we must have $x(r_{i-1}) < x(r_{i+1})$. For the sake of contradiction, assume otherwise. This means we have $x(r_i) < x(r_{i+1}) < x(r_{i-1})$. So we can use the result of Lemma 23 with parameters being $i_1 = i + 1$, $i_2 = i$, $i_3 = i - 1$ and following the points in reverse order of orientation, i.e. $r_{i+1} \to r_i \to r_{i-1}$ (this is the mirrored setting of Lemma 23). This implies r_{i+1} must be a descending reflection in the reverse orientation, which means it must be ascending in the original orientation (that ravels r_i to r_{i+1}). But we assumed r_{i+1} is descending. This contradicts Lemma 23 and proves our initial claim that $x(r_{i-1}) < x(r_{i+1})$. (see Figure 18). Thus, $x(r_i) < x(r_{i-1}) < x(r_{i+1})$.

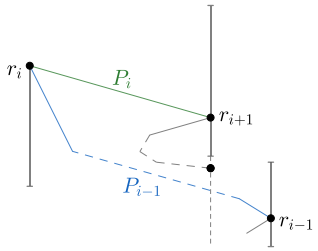


FIGURE 18. If between r_i and r_{i+1} one is ascending and the other is descending, then r_{i-1} (or r_{i+2}) must have an x-coordinate between $x(r_i)$ and $x(r_{i+1})$

Let s_j be the segment of the instance that r_{i-1} lies on. Once again, using Lemma 23, we get that r_{i-1} is an ascending reflection and s_j is a bottom segment (see Figure 19).

According to Lemma 21, we get that the subpath of P from r_1 to r_{i-1} (since it contains the lower leg of r_{i-1} due to it being an ascending reflection) can't reach to the right of s_j.

We will show that the subpath of P from r_{i+1} to r_k will not reach to the left of s_j either. This implies no vertical line can at the same time cross the first part that ends at r_i and the other parts starting at r_{i+1} onward. Repeating this argument implies no vertical line can intersect two parts as wanted. Consider the area surrounded by the line $x = x(s_j)$ and $P_{i-1} \cup P_i$, and refer to it by A_j. Now consider the subpath of P from r_{i+1} to r_k and refer to it as $\mathcal{P}_{i+1}$. Similar to the proof of Lemma 21, $\mathcal{P}_{i+1}$ can't enter A_j, because in order to exit from A_j, it has to reflect at some point inside A_j. But for such a reflection point to exists, there has to be a segment containing it, and that segment will intersect with P_{i-1} or P_i, which contradicts Lemma 7. So we conclude that if $\mathcal{P}_{i+1}$ were to go to the left of s_j, it has to do so from outside of A_j, i.e. from above P_i (since P_i is the upper hull of A_j).

Take two cases based on whether the segment $s_{j'}$ that contains r_{i+1} is a top segment or a bottom segment:

- $s_{j'}$ **is a top segment**:
  The area of S_τ is cut into two parts by $P_{i-1} \cup P_i \cup s_j \cup s_{j'}$. Since r_{i+1} is a descending reflection, then the lower leg of r_{i+1} is in the same part as the bottom tip of $s_{j'}$; refer to this part by A_1 and let A_2 be the other area. Since $\mathcal{P}_{i+1}$ includes this leg, it means that if $\mathcal{P}_{i+1}$ is going to reach to the left of s_j, it has to reach from A_1 to A_2. This would require it to either intersect with P_i or with $s_{j'}$. The former isn't possible because it would make OPT self-crossing, and the latter isn't possible because of Lemma 7.
- $s_{j'}$ **is a bottom segment**:
  The lower leg of r_{i+1} is in the area $A_{j'}$ surrounded by $P_{i-1} \cup P_i \cup s_j \cup s_{j'}$. Since we mentioned $\mathcal{P}_{i+1}$ can't reach inside of A_j, then it needs to exit $A_{j'}$ and go over P_i. This means $\mathcal{P}_{i+1}$ has to either intersect with P_i or $s_{j'}$, which gives us the same contradictions as above.

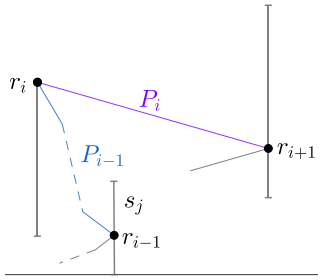


FIGURE 19. If r_i is an ascending reflection and r_{i+1} is a descending reflection, then r_{i-1} must be an ascending reflection lying on a bottom segment. r_i and r_{i+1} can be either on top or bottom segments

So we conclude that there is no vertical line Γ that intersects both subpath $\mathcal{P}_i^- = \bigcup_{u=1}^{i} P_u$ and subpath $\mathcal{P}_{i+1} = \bigcup_{u=i+1}^{k} P_u$. Thus, $r_1, \ldots, r_i$ gives us a partition as desired. By continuing this process for the rest of the reflections, we get that no vertical line can intersect two parts. Since the path between two conseutive parts (last reflection of one part and the first reflection of the next part) has shadow 1, this completes the proof of the last part of Lemma 8. □

4.2. **Properties of a Near-Optimum Solution.** In this section we start with some lemmas that are needed to prove major Lemmas 9, 10, and 11.

Lemma 25. *Let $\mathcal{R} = r_1, r_2, \ldots, r_k$ denote the reflection points for any zig-zag or sink in a strip S_τ where $k \geq 3$. Without loss of generality, assume r_1 is on a bottom segment and is an ascending left reflection point. Then:*

- *If $\mathcal{R}$ is a zig-zag, then $x(r_1) < x(r_3) < \cdots < x(r_{2i-1}) < \ldots$ and $x(r_2) < x(r_4) < \cdots < x(r_{2i}) \ldots$. All the inequalities hold in the other direction if r_1 is a right reflection point.*
- *If $\mathcal{R}$ is a sink then $x(r_1) < x(r_3) < \cdots < x(r_{2i-1}) < \ldots$ and $x(r_2) > x(r_4) > \cdots > x(r_{2i}) \ldots$. Again, all the inequalities hold in the other direction if r_1 is a right reflection point.*

Proof. Assume $r_1, \ldots, r_k$ lie on segments $s_{i_1}, s_{i_2}, \ldots, s_{i_k}$, respectively. Also, let's denote the path (following the orientation of OPT) from r_m to r_{m+1} by P_m. By definition, all P_m's are monotone in the x-coordinates (see Lemma 3).

First, consider the case that $\mathcal{R}$ is a zig-zag. Since r_1 is a left reflection and s_{i_1} is a bottom segment, and all reflection points are ascending, it means r_2 is a right reflection to the left of r_1 (because of Lemma 6), and s_{i_2} is a top segment. This implies P_1 is a decreasing path in the x-coordinate. Once again using Lemma 6, since r_2 is a right reflection, we have $x(r_3) > x(r_2)$. We claim that $x(r_3) > x(r_1)$. If this is not the case, then we have $x(r_2) < x(r_3) < x(r_1)$. Using Lemma 23 for parameters $r_{i_1} = r_3, r_{i_2} = r_2$, and $r_{i_3} = r_1$ in the order $r_3 \to r_2 \to r_1$ (which makes these reflections descending), implies that s_{i_3} must be a top segment, which is a contradiction. So we get $x(r_3) > x(r_1)$. Analogous argument shows that we must have $x(r_2) < x(r_4)$. Iteratively applying this argument establishes the inequalities.

Now consider the case that $\mathcal{R}$ is a sink. The argument is very similar to the case of zig-zag. Note that in this case, all the segments $s_{i_1}, \ldots, s_{i_k}$ are now bottom segments, all the reflection points are ascending, and they must alternate between left and right reflection points. Since r_1 is a left reflection, r_2 is a right reflection with $x(r_2) < x(r_1)$ (due to Lemma 6). We again have $x(r_3) > x(r_2)$ because of Lemma 6. Again, if we have $x(r_3) < x(r_1)$, then we have $x(r_2) < x(r_3) < x(r_2)$. Using Lemma 23 for parameters $r_{i_1} = r_3, r_{i_2} = r_2$, and $r_{i_3} = r_1$ in the order $r_3 \to r_2 \to r_1$ (which means the reflections are descending) implies s_{i_3} is a top segment, a contradiction. Thus, we get $x(r_3) > x(r_1)$. A similar argument shows that $x(r_2) > x(r_4)$, otherwise by an application of Lemma 6, s_{i_4} must be a top segment, which contradicts the assumption of a sink. By iteratively applying the same argument, we obtain the inequalities stated. □

Lemma 26. *If $p_j, p_{j'}$ are consecutive reflection points in OPT, and both are pure reflections and all the other points of OPT in between them (if any) are straight points, then either both $p_j, p_{j'}$ are ascending or both are descending.*

Proof. By way of contradiction, suppose p_j is ascending and $p_{j'}$ is descending. Note that one is a left reflection and the other is a right reflection (as reflection points must alternate). Suppose p_j is a point on segment s_i, and $p_{j'}$ is on segment $s_{i'}$. From the assumption, the path from p_j to $p_{j'}$ is a straight line. Let ℓ_j, ℓ_{j+1} be the two legs incident to p_j and $\ell_{j'}, \ell_{j'+1}$ be the two legs incident to $p_{j'}$. From the definition of pure reflection, we need to have the angle between ℓ_j and s_i and the angle between ℓ_{j+1} and s_i be the same, and the angle between $\ell_{j'}$ and $s_{i'}$ and the angle between $\ell_{j'+1}$ and $s_{i'}$ be the same. The only way this is possible is when $\ell_j, \ell_{j+1}, \ell_{j'+1}$ are all horizontal but this means OPT is self-crossing. This contradiction yeilds the result of the lemma. □

Lemma 27. *Let $\mathcal{R} = r_0, r_1, \ldots, r_k$ be any sequence of reflections that form a sink or zig-zag in a strip S_τ. For $1 \le j \le k$ let P_j be the subpath of $\mathcal{R}$ between r_{j-1} to r_j. Let $\mathcal{P} = \{P_1, P_2, \ldots, P_k\}$. Take any vertical line Γ and let $P_\Gamma = \{P_{j_1}, P_{j_2}, \ldots, P_{j_m}\}$ $(j_1 < j_2 < \cdots < j_m)$ be the maximal subset of $\mathcal{P}$ that each $P_j \in P_\Gamma$ intersect with Γ. Then P_Γ must be a consecutive subset of $\mathcal{P}$. In other words, $P_\Gamma = \{P_{j_1}, P_{j_1+1}, P_{j_1+2}, \ldots, P_{j_1+m-1}\}$.*

Proof. We say a reflection r_j is *included* in P_Γ if $P_j \in P_\Gamma$ or $P_{j+1} \in P_\Gamma$. We prove the following claim to use throughout this proof:

Claim 2. *There are no included left reflections to the left of Γ (and similarly no included right reflections to the right of it).*

Proof of Claim. Assume the contrary, that there is some left reflection r_{j_i} included in P_Γ that is to the left of Γ. Without loss of generality, assume that $P_{j_i} \in P_\Gamma$. Similar to the proof of Lemma 4, the points on P_{j_i} are monotone in the x-coordinate. This means the path from r_{j_i} on P_{j_i}, is decreasing in the x-coordinate (because r_{j_i} has its legs facing left), implying P_{j_i} is completely to the left of r_{j_i}. Since Γ is to the right of r_{j_i}, this means that P_{j_i} can't intersect with Γ, contradicting the assumption that $P_{j_i} \in P_\Gamma$. This proves the claim. □

Now back to the statement of the lemma; Without loss of generality, assume that reflections in $\mathcal{R}$ are all ascending. For the sake of contradiction, assume that there is an index $1 \le a < m$ for which P_{j_a} and $P_{j_{a+1}}$ aren't consecutive. This means $j_a < j_{a+1} - 1$, and so we conclude that the subpath $P' = \bigcup_{j'=j_a+1}^{j_{a+1}-1} P_{j'}$ of $\mathcal{R}$ from r_{j_a} to $r_{j_{a+1}} - 1$ is on one side of Γ (or else there will be another $P_{j'} \in P_\Gamma$ with $j_a < j' < j_{a+1}$). So there is at least one reflection point from $\mathcal{R}$ that is in P'. Let r_i be the first reflection on P' after r_{j_a}. Without loss of generality, assume that r_{j_a} (and therefore the entirety of P') is on the right side of Γ. So r_{j_a} is a left reflection because of Claim 2. By Lemma 6, both r_i and r_{j_a-1} (the reflection in $\mathcal{R}$ before r_{j_a}) are right reflections.

Let r_q be the end-point of $P_{j_{a+1}}$ that is to the left of Γ (either $r_q = r_{j_{a+1}}$ or $r_q = r_{j_{a+1}-1}$). Once again, using Claim 2, we get that r_q is a right reflection. So we have three right reflections r_{j_a-1}, r_i, and r_q such that $x(r_i) \ge x(\Gamma) \ge \{x(r_{j_a-1}), x(r_q)\}$ and the order they're visited in $\mathcal{R}$ is r_{j_a-1}, then r_i, and then r_q. According to Lemma 25, based on whether $\mathcal{R}$ is a sink or a zig-zag, we either must have $x(r_{j_a-1}) < x(r_i) < x(r_q)$ or the reversed inequality; which neither are the case here. This contradiction implies the statement of the lemma. □

4.2.1. *Proof of Lemma 9: Bounding the Shadow of each Sink/Zig-zag.* This subsection is dedicated to the proof of Lemma 9.

Let $\sigma = \lceil 1/\varepsilon \rceil + 1$ and consider any loop or ladder $P \in \mathrm{OPT}_\tau$ and let $\mathcal{R}$ be an arbitrary zig-zag/sink along P with shadow larger than σ at some vertical line $x = x_0$. Without loss of generality, assume that following the orientation of OPT along P, reflection points on $\mathcal{R}$ are ascending. Suppose that the subpath of P following reflection points $r_j, r_{j+1}, \ldots, r_k$ of $\mathcal{R}$ is crossing $x = x_0$ (note that, using Lemma 27, the reflection points must be consecutive). Let this subpath of P starting at r_j and ending at r_k be $\mathcal{R}'$ and let $s_{a_j}, s_{a_{j+1}}, \ldots, s_{a_k}$ denote the segments that contain reflections $r_j, r_{j+1}, \ldots, r_{j+k}$, respectively. Also let P_i (for $1 \le i \le k-j$) be the subpath of $\mathcal{R}'$ from r_{j+i-1} to r_{j+i}. Note that there might be several straight points or break points between $r_{j'}, r_{j'+1}$ on P (for each j'); the segments of these points are all covered by the shadow one path (due to Lemma 3) from $r_{j'}$ to $r_{j'+1}$. According to Lemma 5, since all reflections are ascending, if $m_1 < m_2$, then P_{m_1} is below P_{m_2} (in the range that P_{m_1} is defined on the x-axis). So this specifically implies P_1 is below any other P_m (in the range that P_1 is defined), and similarly, P_k is above any other P_m (in the range that P_k is defined). Note that $\mathcal{R}'$ is part of a zig-zag/sink itself (the only difference with the definition of zig-zag/sink is that $\mathcal{R}'$ is no longer necessarily maximal in OPT_τ). Also, note that for each path P_i, the x value of the points it visits between the two reflection points r_{j+i-1} to r_{j+i} are monotone increasing or decreasing (see Lemma 3). Let Ψ denote the cost of legs of $\mathcal{R}'$. It follows that $r_j, r_{j+2}, r_{j+4}, \ldots$ are on one side of x_0 (say to the right) and $r_{j+1}, r_{j+3}, \ldots$ are on the other side (say left of x_0). Since the number of reflections to the right of $x = x_0$ differs from the number of reflections to the left of $x = x_0$ by at most 1, then on each side of $x = x_0$ we have at least $(\sigma - 1)/2 = \lceil 1/2\varepsilon \rceil$ reflections. Let $\sigma' = \lceil 1/2\varepsilon \rceil$. The idea of the proof is to show that aside from the $2\sigma'$ reflections at the end of $\mathcal{R}'$ (i.e. the last $2\sigma'$ paths P_j), we can replace the paths between the rest of the reflection points so that it reduces the shadow of the entire $\mathcal{R}'$ to $O(1/\varepsilon)$ while increasing the cost of the path by at most $O(\varepsilon \cdot \Psi)$.

Note that using Lemma 3, each subpath P_j of $\mathcal{R}'$ is between two consecutive reflection points and so has a shadow of 1. This implies that $P_{k-2\sigma'} \cup \ldots \cup P_{k-1}$ has a shadow of $O(\frac{1}{\varepsilon})$ as it has $2\sigma'$ consecutive reflections. We replace the rest of $\mathcal{R}$ (as we describe below) with a new path of a shadow of $O(1)$; this will yield the result of the lemma.

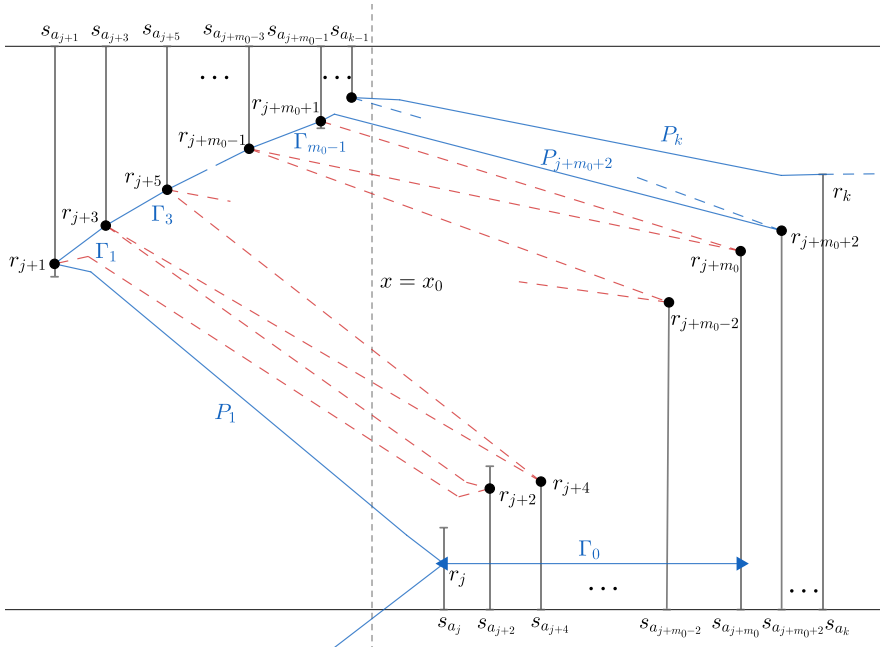


FIGURE 20. Alternative path for a zig-zag; The red dashed parts are discarded. There are further details about Γ_m's that are explained throughout the proof of Lemma 9

When $\mathcal{R}'$ is a part of a zig-zag. Without loss of generality, assume that $s_{a_j}, s_{a_{j+2}}, \ldots$ are bottom segments and to the right of $x = x_0$, and consequently, $s_{a_{j+1}}, s_{a_{j+3}}, \ldots$ are top segments and to the left of $x = x_0$. Let

d_m for $m = 0, 1, \ldots, k-j$ denote $|x(r_{j+m}) - x_0|$. Using Lemma 25, we have $d_0 < d_2 < \cdots$ and $d_1 > d_3 > \cdots$. Let's focus on the right side of $x = x_0$ (where the bottom segments are), so $r_j, r_{j+2}, \ldots, r_{j+2q}$ are all the reflections of $\mathcal{R}'$ on this side where $q \geq \sigma' - 1$. We claim that except for at most the σ' largest values in $d_2, d_4, d_6, \ldots$, all other values of d_{2m}'s are at most $\varepsilon \cdot \Psi$ and this is done by an averaging argument. More specifically, we show that the largest integer $m_0 \in \{0, 2, 4, \ldots, 2q\}$ for which we have $d_{j+m_0} \leq \varepsilon \cdot \Psi$, has value $m_0 \geq 2(q-(\sigma'-1))$. To see why this is the case, assume otherwise, that for all even integers $m \geq 2(q-(\sigma'-1))$, we have $d_{j+m} > \varepsilon \cdot \Psi$. Adding these inequalities for $m = 2(q-(\sigma'-1)),\ 2(q-(\sigma'-2)),\ \ldots,\ 2q$ gives us

$$\begin{aligned} d_{j+2(q-(\sigma'-1))} + d_{j+2(q-(\sigma'-2))} + \cdots + d_{j+2(q)} &> \sigma' \cdot (\varepsilon \cdot \Psi) \\ &= \lceil 1/2\varepsilon \rceil \cdot \varepsilon \cdot \Psi \\ &\geq \Psi/2, \end{aligned}$$

which clearly isn't possible, due to $2\sum_{m \in \{2(q-(\sigma'-1)),\ldots,2q\}} d_{j+m} \leq \Psi$; this inequality holds because paths P_{j+m}, P_{j+m+1} ($m \in \{2(q-(\sigma'-1)), \ldots, 2q\}$) have to travel the x-distance from x_0 to $s_{a_{j+m}}$ to the reflection points r_{j+m}, and all these paths are part of $\mathcal{R}'$. This contradiction shows our initial claim, that for some $m_0 \geq 2(q-(\sigma'-1))$, we have all of $d_j, d_{j+2}, \ldots, d_{j+m_0} \leq \varepsilon \cdot \Psi$.

We are going to change $\mathcal{R}'$ from r_j up to r_{j+m_0}, but keep P_{j+m_0+1} and after; this change will result in another feasible solution with an $O(1)$ shadow up to r_{j+m_0}, and cost increase will be at most $O(\varepsilon \cdot \Psi)$. Our modification of $\mathcal{R}'$ is informally as follows (skipping some details to be explained soon). Starting at r_j instead of following P_1 to r_{j+1}, we first travel horizontally to the right until we hit $s_{a_{j+m_0}}$ (the bottom segment which r_{j+m_0} is located on), and travel back to r_j. Let's call this horizontal back and forth subpath Γ. This subpath Γ will ensure that all the bottom segments that $\mathcal{R}'$ covers between $x = x_0$ and $s_{a_{j+m_0}}$ are covered (we may need to deviate from Γ further down if $\mathcal{R}'$ goes further below Γ at some point; will formalize this soon). The shadow of Γ will easily be shown to be 2. Then from r_j, we follow P_1 and go to r_{j+1} which is the left-most reflection on a top segment (to the left of $x = x_0$). Now instead of following P_2 to go to r_{j+2} and then P_3 to go to r_{j+3}, we go straight from r_{j+1} to r_{j+3} (with some little details skipped here), then to r_{j+5} and so on until we get to r_{j+m_0+1}, and from there we follow $\mathcal{R}'$. One observation is that the shadow of the new path from r_{j+1} to r_{j+m_0+1} is also 1 since it won't have any reflection points. The rest of the path from r_{j+m_0+1} to r_k that follows $\mathcal{R}'$ has at most $O(\sigma')$ reflection points and hence the shadow is $O(1/\varepsilon)$. We show that the new path hits all the segments $\mathcal{R}'$ were hitting; and so we still have a feasible solution where the overall increase in the cost is at most $O(d_{j+m_0})$, which is bounded by $O(\varepsilon \cdot \Psi)$. Hence, we find a modification of the path $\mathcal{R}'$ with shadow bounded by $O(1/\varepsilon)$, and cost increase is at most $O(\varepsilon \cdot \Psi)$. Note that any bottom segment (if any) to the left of $x = x_0$ that was covered by $\mathcal{R}'$, must intersect P_1; as P_1 is below the rest of $\mathcal{R}'$ to the left of $x = x_0$. Thus, any bottom segment to the left of x_0 that is covered by any of the $P_{b>1}$, is also covered by P_1. There are some details missing in this informal description that are explained below.

We will introduce a new subpath Γ_0, responsible for covering all bottom segments in $\mathcal{R}'$ to the right of $x = x_0$ until s_{j+m_0}; and we introduce a collection of subpaths Γ_m for odd m in $\{1, \ldots, m_0\}$ for covering the top segments to the left of $x = x_0$. All of Γ_m's, will have a shadow of 1. We ensure that any bottom segment hit by $\mathcal{R}'$ between s_{j_0} and s_{j+m_0}, is also hit by Γ_0 between $x = x(s_j)$ and $x = x(s_{j+m_0})$; and also any top segments that $\mathcal{R}'$ was hitting in the range that each Γ_m is defined, is hit by Γ_m, for $m \geq 1$.

Consider the horizontal line $y = y(r_j)$ from s_{a_j} to $s_{a_{j+m_0}}$. Refer to this horizontal portion as Γ. For reflection points $r_j, r_{j+2}, \ldots, r_{j+m_0}$ (on bottom segments $s_{a_j}, s_{a_{j+2}}, \ldots, s_{a_{j+m_0}}$), the two paths that contain a leg incident to r_{j+m} are P_{j+m} and P_{j+m+1} for each $0 \leq m \leq m_0$. Recall that using Lemma 5, P_{j+m} is below P_{j+m+1} between s_{a_j+m-1} and s_{a_j+m}. Consider the area A_Γ of the strip bounded by $\Gamma \cup s_{a_j} \cup s_{a_{j+m_0}}$. Then $\mathcal{R}' \cap A_\Gamma$ are (possibly empty) subpaths that start and end at Γ. These subpaths form the lower-envelope of $\mathcal{R}' \cup \Gamma$ in A_Γ (for e.g. in Figure 21, paths P_4, P_5 that reach r_{j+4}, cross Γ at points q_1^4, q_2^4.).

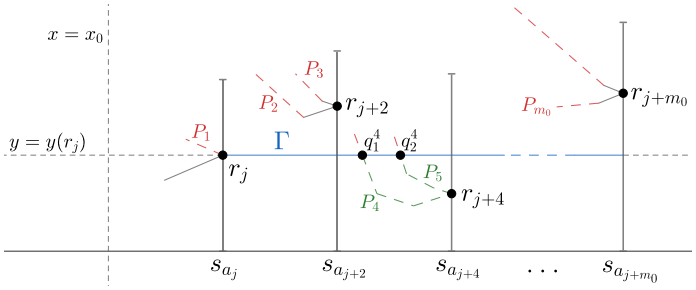


FIGURE 21. Γ_0 is the lower envelope of the blue line (the segment Γ) together with the green parts (portions of OPT_τ that go below Γ)

We define Γ_0 to be a path starting at r_j that travels right along Γ and the lower envelope of $\mathcal{R}'$ in this area, i.e. whenever travelling right on Γ, if we arrive at an intersection of $\mathcal{R}'$ with Γ (say a path P_{j+m}) then we travel along P_{j+m} inside A_Γ until we hit back at Γ, and then continue travelling right. For instance, in Figure 21, when travelling on Γ from r_j to right, once we arrive at q_1^4, we follow P_4 to r_{j+4}, then follow P_5 to q_2^4, and then continue right on Γ. Once we arrive at $s_{a_{j+m_0}}$, we travel Γ horizontally back to r_j. The length of Γ_0 can be bounded by the length of $\mathcal{R}' \cap A_\Gamma$ plus $2d_{j+m_0} \leq 2\varepsilon\Psi$. Also, it can be seen that any bottom segment that was covered by $\mathcal{R}'$ in between $x(r_j)$ and $x(r_{j+m_0})$, is covered by Γ_0 (since we travel the lower envelope of $\mathcal{R}' \cup A_\Gamma$ in the range we're defining Γ_0). Any top segment that is covered by $\mathcal{R}'$ within $[x(r_j), x(r_{j+m_0})]$, must be also covered by P_{m_0+1}; as that path is above all other P_m's in the range of $[x_0, x(r_{j+m_0})]$. After travelling Γ_0, we travel along P_1 to r_{j+1}. Now we're going to define Γ_m for odd $1 \leq m \leq m_0$. Each Γ_m goes from r_{j+m} to r_{j+m+2} until we arrive at r_{j+m_0+1}; after which we follow along $\mathcal{R}'$ (i.e. P_{j+m_0+2}, then P_{j+m_0+3} and so on). Path Γ_1 will replace $P_2 + P_3$, Γ_3 will replace $P_4 + P_5$, and so on. Note that Γ_m's are all to the left of $x = x_0$. For any two reflections r_{j+m} and r_{j+m+2} that lie on top segments $s_{a_{j+m}}$ and $s_{a_{j+m+2}}$, let γ_m be the subpaths of $\mathcal{R}'$ restricted to the area of the strip cut by segment $r_{j+m}r_{j+m+2}$ and $s_{a_{j+m}}$ and $s_{a_{j+m+2}}$ (i.e. the area between $s_{a_{j+m}}$ and $s_{a_{j+m+2}}$ and above $r_{j+m}r_{j+m+2}$). Path Γ_m is obtained by starting at r_{j+m} and following line $r_{j+m}r_{j+m+2}$ and whenever we hit $\mathcal{R}'$, i.e. a subpath of γ_m (see Figure 22) we follow that subpath until we arrive back to $r_{j+m}r_{j+m+2}$ again; we continue until we reach r_{j+m+2}. In other words, we follow the upper envelope of $r_{j+m}r_{j+m+2} \cup \mathcal{R}'$ between r_{j+m} and r_{j+m+2}. If we define $q_1^m, \ldots, q_{2i_m}^m$ to be the intersections of $\mathcal{R}'$ with Γ_m, ordered in the direction of $r_{j+m} \to r_{j+m+2}$,

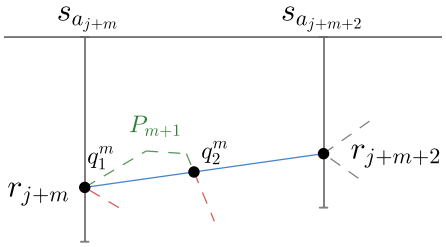


FIGURE 22. Γ_m is travelling the upper envelope of the blue line (the segment $r_{j+m}\, r_{j+m+2}$) and the green parts (portions of OPT_τ that go above $r_{j+m}\, r_{j+m+2}$)

then if $\mathcal{R}'$ intersects with $r_{j+m}r_{j+m+2}$ and goes above it, it has to be at a point q_u^m where u is odd, and otherwise u has to be even.

Overall, we have changed the subpaths of $\mathcal{R}'$ from r_j to r_{j+m_0} as follows (see Figure 20):

- Follow Γ from r_j towards $s_{a_{j+m_0}}$, such that every time an intersection point with $\mathcal{R}'$ (say point q_{2u-1}^0) is reached, then follow along $\mathcal{R}'$ until the next intersection of $\mathcal{R}'$ with Γ (say point q_{2u}^0) is reached; then continue along Γ. Repeat this process until we reach $s_{a_{j+m_0}}$, then follow along Γ from right to left directly back to r_j; this is subpath Γ_0
- From r_j, follow P_1 to reach r_{j+1}.
- From r_{j+m} (initially $m = 1$), follow Γ_m similar to the first step; meaning follow the segment $r_{j+m}\, r_{j+m+2}$, and when an intersection point q_{2u-1}^m with $\mathcal{R}'$ is reached, follow $\mathcal{R}'$ instead, until

you reach the next intersection point q_{2u}^m on $r_{j+m}r_{j+m+2}$. Repeat this process (for $m = 1, 3, \dots$) until r_{j+m_0+1} is reached.

- From r_{j+m_0+1}, follow P_{j+m_0+2} and the rest of $\mathcal{R}'$ to the end.

First, we show that we still have a feasible solution, i.e. every segment that $\mathcal{R}'$ used to cover, will have an intersection with the new solution. To see why this is the case, first note that Γ_0 by definition is always on or below $\mathcal{R}'$ in the ranges it's defined; this means that (using Observation 2) Γ_0 covers any bottom segment that $\mathcal{R}'$ covers between s_{a_j} to $s_{a_{j+m_0}}$. Also, all the top segments in this range will be intersecting the path P_{m_0+1}, since P_{m_0+1} is above all $P_{\leq m_0}$ in this range. Similarly, each Γ_m is on or above $\mathcal{R}'$ in the range $[x(r_{j+m}), x(r_{j+m+2})]$, meaning they cover all the top segments that $\mathcal{R}'$ used to cover between $s_{a_{j+1}}$ to $s_{a_{j+m_0+1}}$. Also, any bottom segment that was covered in this range is covered by P_1.

Next, note that from r_j to r_{j+m_0+1}, the shadow is at most 3. That is because shadow of Γ_0 is 2, shadow of each Γ_m $(1 \leq m)$ is 1, and shadow of P_1 is 1.

Now we show the new solution has an additional cost of at most $3\varepsilon\Psi$. All parts of Γ_0 and the rest of Γ_m's that used portions of $\mathcal{R}'$ can be charged onto $\mathcal{R}'$ itself. So we only have to properly charge the line segment Γ along with its duplicate (part of Γ_0) and line segments $r_{j+m}r_{j+m+2}$ (part of Γ_m). We know that $||\Gamma|| = x(r_{j+m_0}) - x(r_j) < x(r_{j+m_0}) - x_0 = d_{m_0} \leq \varepsilon \cdot \Psi$. So we pay at most $2\varepsilon\Psi$ extra (compared to OPT_τ) for travelling Γ_0. We consider one additional copy of Γ for the extra cost we pay elsewhere in Γ_m $(m \geq 1)$, and we are going to use this for our charging scheme. So at the end, the total extra cost is going to be bounded by $3\varepsilon\Psi$.

For each two reflections r_{j+m} and r_{j+m+2} that lie on top segments, note that $\mathcal{R}'$ had two subpaths paths P_{j+m+1}, P_{j+m+2} whose concatenation makes a path from r_{j+m} to r_{j+m+2}; but now, it is possible that some portions of P_{j+m+1}, P_{j+m+2} are used in Γ_0 during our alternate solution (those that belonged to A_Γ). But having that additional copy of Γ that we accounted for, we can use it to short-cut the missing parts of $P_{j+m+1} \cup P_{j+m+2}$ to again make a path from r_{j+m} to r_{j+m+2}. Overall, the total length of $P_1 + \sum_{m\geq 0} \Gamma_m$ that is replacing $P_1 + P_2 + \ldots + P_{j+m_0+1}$ is at most $3\varepsilon\Psi$ larger than length of $P_1 + P_2 + \ldots + P_{j+m_0+1}$. Thus, we conclude the lemma for the case of zig-zags.

When $\mathcal{R}'$ is a part of a sink. The proof is analogous to the case of zig-zags. Without loss of generality, assume all reflections in $\mathcal{R}'$ are on bottom segments. Define $d_m = |x(r_{j+m}) - x_0|$ like before. If $r_j, r_{j+2}, \dots, r_{j+2q}$ are all the reflections to the right of $x = x_0$, then with the same arguments as the case of zig-zags, we will find an integer $m_0 \geq 2(q - (\sigma' - 1))$ $(\sigma' = \lceil \frac{1}{2\varepsilon} \rceil)$ for which we have $d_{m_0} \leq \varepsilon \cdot \Psi$.

We will replace the subpath of $\mathcal{R}'$ from r_j to r_{j+m_0} in the same fashion as before. Let Γ be the segment on the line $y = y(r_j)$ in the range $[x(r_j), x(r_{j+m_0})]$. Define Γ_0 to be the union of Γ with the portions of $\mathcal{R}'$ that go below it. Define each Γ_m for a reflection r_{j+m} to the left of $x = x_0$ to be the union of segment $r_{j+m}\ r_{j+m+2}$ with the portions of $\mathcal{R}'$ that go below it.

The same arguments as before hold, that each of Γ or $r_{j+m}\ r_{j+m+2}$ that we defined above, will have an even number of intersections with $\mathcal{R}'$. Define the new path between these reflections in the same way as we did for zig-zags.

The cost arguments still hold, implying that the new path has an additional cost of $O(\varepsilon \cdot \Psi)$. Also, the new path is always on or below $\mathcal{R}'$, so it covers all the bottom segments that $\mathcal{R}'$ used to cover previously. But one can see that P_{m_0} is above all of $\mathcal{R}'$ in the path between r_j to r_{j+m_0}. Thus, P_{m_0} alone will cover all top segments that $\mathcal{R}'$ used to cover. Once again, the new solution has a shadow of at most 3 in the subpath between r_j and r_{j+m_0} and shadow $O(1/\varepsilon)$ afterwards. This concludes the proof for the case of sinks and the proof of Lemma 9. □

4.2.2. *Proof of Lemma 10: Bounding the Size of Pure Reflection Sequences.* In this subsection, we will prove Lemma 10.

We prove the lemma by showing how to change each ladder or loop (i.e. any path of OPT_τ that starts and ends on one of the cover-lines) so that the size of each pure reflection sub-sequence is bounded without increasing the cost by more than $(1 + \varepsilon)$ factor. Consider any loop or ladder $P \in \text{OPT}_\tau$ and any maximal pure reflection sequence $P' = r_0, r_2, \dots, r_k$ in P where $k > \frac{1}{\epsilon}$. Let Ψ be the length of subpath of OPT_τ from r_0 to r_k. We show how we can modify this subpath to another one whose length is at most $(1 + O(\varepsilon))\Psi$ such that the length of each pure reflection sub-sequence is bounded by $O(1/\varepsilon)$. Note that using Lemma 26, all r_i's are ascending or all are descending. This also implies that the y-coordinates of r_i's are monotone. i.e. either $y(r_0) \leq y(r_1) \leq \cdots \leq y(r_k)$ or the other way around. Without loss of generality, assume it is

the former case and so all are ascending reflection points. Proof of Lemma 8, shows if we have a maximal monotone (i.e. all ascending or all descending) sequence of reflection points, then it consists of at most a sink followed by a zig-zag, followed by a sink. Therefore, it suffices to bound the size of pure reflection sequence in a single sink or a zig-zag alone as a function of $1/\varepsilon$. So let's assume all r_i's form a single sink or all form a single zig-zag.

Recall from the definition of pure reflection sequence that there might be straight points in P between two consecutive reflection points. For any reflection point r_i on a segment s, let d_i^+ and d_i^- be the distances of r_i to the top and bottom tips of s, respectively. By the definition of a pure reflection sequence, $d_i^- > 0$ and $d_i^+ > 0$ for all $0 \le i \le k$ (because the reflections are not at the tips). With $\sigma = \lceil \frac{1}{\epsilon} \rceil$, we break P' into m subpaths $G_1, \ldots, G_m$ where G_j is the subpath of P' from $r_{j-1\sigma}$ to $r_{j\sigma}$, except that the last group ends at r_k. Note that the concatenation of these paths is P', and each subpath has at most $\sigma+1$ reflections. Consider any group G_j and let $\mathcal{G}_j$ be the cost of the legs of P between the reflection points of G_j and let D_j be the smallest value among minimum of d_a^+, d_a^- among all reflection points $r_a \in G_j$, i.e. $D_j = \min_{(j-1)\sigma \le a \le j\sigma}\{d_a^+, d_a^-\}$.

Claim 3. *For each* $1 \le j \le m$: $D_j \le \frac{2\varepsilon}{1-\varepsilon} \cdot \mathcal{G}_j$.

Proof of Claim. For simplicity of notation of indices, we prove this for $j = 1$, i.e. $G_1 = r_0, \ldots, r_\sigma$. As mentioned above, it suffices to show the claim for G_1 being part of a sink or a zig-zag.

• G_1 **is part of a sink**:

Without loss of generality, assume all reflections in G_1 are on bottom segments. Consider the three consecutive reflection points r_0, r_1, and r_2. Using Lemma 7, we get that subpath $r_1 \to r_2$ (which, according to the definition of pure reflection sequence, is a straight line) does not intersect with the segment containing r_0. Since we assumed the y-coordinates of r_i's are increasing, this implies that r_1r_2 and consequently r_2 lie above the segment containing r_0. This along with triangle inequality yields us $d_0^+ \le y(r_2) - y(r_0) \le ||r_2r_0|| \le ||r_0r_1|| + ||r_1r_2||$. With the same argument, we get $d_i^+ \le y(r_{i+2}) - y(r_i) \le ||r_ir_{i+1}|| + ||r_{i+1}r_{i+2}||$ for all $0 \le i \le \sigma - 2$ (see Figure 23). Considering these inequalities for different r_i's ($1 \le i \le \sigma - 2$) and summing them up for all r_i's in group G_1, using the fact that $D_1 \le d_i^+$, we obtain

$$(\sigma - 1) \cdot D_1 \le \sum_{i=1}^{\sigma-2} d_i^+ \le \sum_{i=1}^{\sigma-2} (||r_ir_{i+1}|| + ||r_{i+1}r_{i+2}||) \le 2 \cdot \mathcal{G}_1$$

$$\implies D_1 \le \frac{2}{\sigma - 1} \cdot \mathcal{G}_1.$$

This implies in a sink, $D_1 \le \frac{2}{1/\varepsilon - 1} \cdot \mathcal{G}_1 = \frac{2\varepsilon}{1-\varepsilon} \cdot \mathcal{G}_1$ and in general, $D_j \le \frac{2\varepsilon}{1-\varepsilon} \cdot \mathcal{G}_j$ for all $1 \le j \le m$.

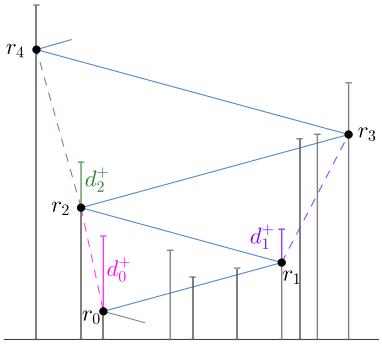


FIGURE 23. Each d_i^+ ($i \le \sigma - 2$) can be charged into the lines reaching the next two reflections

• G_1 **is part of a zig-zag**:

The inequalities are almost analogous, but there are two of them. Without loss of generality, assume that $r_1, r_3, \ldots$ are on bottom segments, and therefore $r_0, r_2, r_4, \ldots$ are on top segments. We give inequalities for d_i^+ on bottom segments, and for d_i^- on top segments. For $i = 1, 3, \ldots$ with $i \le \sigma - 2$, similar to the case

of G_1 being a sink, we have $d_i^+ \leq y(r_{i+2}) - y(r_i) \leq ||r_i r_{i+1}|| + ||r_{i+1} r_{i+2}||$. For $i = 2, 4, 6, \ldots$, we have $d_i^- \leq y(r_i) - y(r_{i-2}) \leq ||r_i r_{i-1}|| + ||r_{i-1} r_{i-2}||$ (see Figure 24).

Now if we add these inequalities (with proper selection between d_i^+ and $d_{i'}^-$) we get

$$\begin{aligned}(\sigma - 1) \cdot D_1 &\leq (d_1^+ + d_3^+ + \cdots) + (d_2^- + d_4^- + \cdots) \\ &\leq \sum_{i \text{ is odd},\ i \geq 1} (||r_i r_{i+1}|| + ||r_{i+1} r_{i+2}||) + \sum_{i \text{ is even},\ i \geq 2} (||r_i r_{i-1}|| + ||r_{i-1} r_{i-2}||) \\ &\leq 2 \cdot G_1 \\ \implies D_1 &\leq \frac{2}{\sigma - 1} \cdot \mathcal{G}_1\end{aligned}$$

And like before, this implies that in a zig-zag, $D_1 \leq \frac{2\varepsilon}{1-\varepsilon} \cdot \mathcal{G}_1$, and in general, $D_j \leq \frac{2\varepsilon}{1-\varepsilon} \cdot \mathcal{G}_j$.

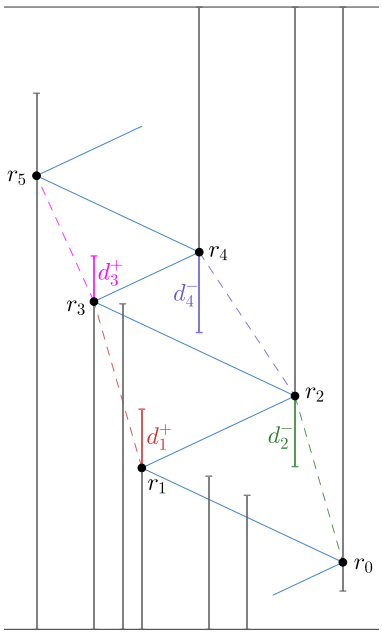


FIGURE 24. d_i^+ $(i \leq \sigma - 2)$ on bottom segments and $d_{i'}^-$ $(i' \geq 2)$ on top segments can be charged into the lines reaching the next/previous two reflections

So we see that the claim holds for loops and ladders. □

Also note that if we have any three consecutive points p, r_j, q on OPT_τ where r_j is a reflection on segment s (with s^t being its top tip), then $||pr_j|| + ||r_j q|| + 2d_j^+ \geq ||ps^t|| + ||s^t q||$ using triangle inequality. Now consider any group G_j $(1 \leq j \leq m)$ and assume r_{j^*} is a reflection point in G_j that lies on the segment s for which $d_{j^*}^+ = D_j$ (if $d_{j^*}^- = D_j$, then consider the bottom tip, s^b instead). Consider the two legs of OPT_τ incident to r_{j^*}, namely ℓ_{j^*-1} and ℓ_{j^*}. Let $\ell_{j^*-1} = p_a r_{j^*}$ and $\ell_{j^*} = r_{j^*} p_b$ where p_a and p_b are points on OPT_τ. Suppose we move r_{j^*} from its current location to s^t, i.e. replace the two legs with $p_a s^t$ and $s^t p_b$. Note that this will remain a feasible solution, as ℓ_{j^*-1}, ℓ_{j^*} have no other intersections with any other segment (as the definition of legs). The new cost is upper bounded by $||p_a r_{j^*}|| + ||r_{j^*} p_b|| + 2d_j^+$, which means the increase is bounded by $2d_j^+ = 2D_i \leq \frac{4\varepsilon}{1-\varepsilon} \cdot \mathcal{G}_i$.

Note that in this new solution in each group G_i, one of the points is moved to be a tip of the segment it lies on. This implies the maximum size of a pure reflection sequence is now bounded by $2\sigma = 2\lfloor 1/\varepsilon \rfloor$ and the total increase in the cost (over all groups) is bounded by $\sum_j \frac{4\varepsilon}{1-\varepsilon} \cdot \mathcal{G}_j = O(\varepsilon \cdot \sum_j \mathcal{G}_j) = O(\varepsilon \Psi)$. This completes the proof of Lemma 10. □

4.2.3. *Proof of Lemma 11: Bounding the Number of Overlapping Loops/Ladders.* Our goal in this subsection is prove Lemma 11, that shows that any given vertical line can intersect with at most $O(1)$ many loops or

ladders of OPT_τ in a strip S_τ. This together with Corollary 2 implies there is a $(1+\varepsilon)$-approximate solution where the shadow in each strip S_τ is $O(1/\varepsilon)$.

We will show that there are at most 12 overlapping loops, and at most 7 overlapping ladders in OPT_τ. Suppose there is a vertical line Γ and a number of loops and ladders are all crossing Γ. We bound the number of loops separately from the number of ladders.

Overlapping Loops. For each of the cover-lines of S_τ, we will show that there are at most 6 overlapping loops that have both their entry points on that cover-line. This will imply that there are at most 12 overlapping loops in total. So from this point onward, let's focus on all overlapping loops on the bottom cover-line. This holds for all the claims and proofs that we introduce in this subsection, unless stated otherwise.

Recall that OPT is not self-crossing, so it cannot have two overlapping cover-line loops. We say a loop L_1 with entry points e_1, o_1 is *nested* over loop L_2 with entry points e_2, o_2 if both e_2, o_2 are between e_1, o_1.

Lemma 28. *Let L_1 and L_2 be any two loops such that L_1 is nested over L_2. Let p_r^2 and p_l^2 be the right-most and left-most points on L_2, respectively. Then, in the range $I = [x(p_l^2), x(p_r^2)]$, L_1 is above L_2. For simplicity, in this case, we say L_1 is above L_2.*

Proof of Lemma 28. For $L_j, j = 1, 2$, let e_j and o_j be its entry points and without loss of generality, assume that $x(e_1) \le x(e_2) \le x(o_2) \le x(o_1)$. So L_1 is a path from e_1 to o_1; meaning it crosses the vertical lines $x = x(e_2)$ and $x = x(o_2)$ at some point. This implies if $\mathcal{L}_1$ is the area of strip S_τ bounded by L_1 and the bottom cover-line, then L_2 is entirely inside $\mathcal{L}_1$. This means if the left-most and right-most points on L_1 are p_l^1 and p_r^1, then $x(p_l^1) \le x(p_l^2)$ and $x(p_r^1) \ge x(p_r^2)$. So we conclude that L_1 is defined in the range $I' = [x(p_l^1), x(p_r^1)]$ and that $I \subseteq I'$. Therefore, in particular, L_1 is defined in the range I and is above L_2. □

Lemma 29. *Suppose L_1 with entry points e_1, o_1 and L_2 with entry points e_2, o_2 are overlapping such that $x(e_1) < x(e_2) < x(o_1)$. Then L_1 must be nested over L_2 and L_2 is a cover-line loop.*

Proof of Lemma 29. If L_1, L_2 are not nested (i.e. $x(e_1) < x(e_2) < x(o_1) < x(o_2)$) and none is a cover-line loop, then they are intersecting inside S_τ, a contradiction. If they are not nested and one (say L_2) is a cover-line loop, then again they are intersecting at one of the entry points. So they must be nested, say $x(e_1) < x(e_2) < x(o_2) < x(o_1)$. Thus, using Lemma 28, L_1 is above L_2; and if L_2 intersects with any top segment, L_1 would already be intersecting with it because of Observation 2. So L_2 should only cover bottom segments, which means it must be a cover-line loop by Lemma 19. □

Using these lemmas, it follows that there are at most 2 overlapping loops with entry points on opposite sides of Γ. Furthermore, if there are two such loops, then one of them is a cover-line loop.

We will finally show that there are at most 2 overlapping loops that have both their entry points on the same side, say left of Γ. This will imply the result of the lemma for loops, because on each of the cover-lines, there are at most 2 loops with entry points on the left of Γ, 2 with entry points on the right, and 2 with entry points on the opposite sides. Between the loops with both entry points to the left of Γ, none can be a cover-line loop because such a loop cannot intersect Γ (Γ needs to be between the two entry points of a cover-line loop). We will show that there will be at most 2 (non-cover-line) overlapping loops with entry points to the left of Γ.

For the sake of contradiction, assume that there are at least 3 loops with entry points on the left of Γ that none are cover-line loops and all cross Γ. Let L_1, L_2, and L_3 be any 3 consecutive loops with this property. Without loss of generality let $x(e_1) \le x(o_1)$, $x(e_2) \le x(o_2)$, and $x(e_3) \le x(o_3)$, and assume an order for the entry points of L_m's, say $x(e_1) \le x(e_2) \le x(e_3)$. We must have $x(e_2) \ge x(o_1)$; or else L_1, L_2 must be nested by Lemma 29, implying L_2 should be a cover-line loop which contradicts the assumption. So we get that $x(e_2) \ge x(o_1)$. Similarly, we have $x(e_3) \ge x(o_2)$. These imply that $e_1, o_1, e_2, o_2, e_3, o_3$ appear in this order on the bottom cover-line. Corollary 3 implies each of L_1, L_2, and L_3 must exclusively cover some top segment. Let r_1, r_2 be the right-most point on L_1, L_2, respectively. Since each L_1, L_2 starts and ends on the left of Γ and travels to the right of Γ, by Lemma 1, the right-most point on each is a reflection point, which implies it must be exclusively covered by using Lemma 7. Let s_{i_1} be the segment that reflection point r_1 lies on, and similarly s_{i_2} the segment for r_2 (see Figure 25).

Lemma 30. *s_{i_1}, s_{i_2} are top segments and $x(s_{i_1}) < x(s_{i_2})$.*

Proof of Lemma 30. By way of contradiction, assume s_{i_1} is a bottom segment. Consider the two subpaths of L_1 between the entry points e_1, o_1 and r_1, let us denote them by $P_r^1 : e_1 \to r_1$ and $P_l^1 : r_1 \to o_2$. L_2

(starting at e_2) is in the region bounded by $P_r^1 \cup s_r^1$ and the bottom cover-line, which means L_1 will intersect any top segment L_2 intersects with (i.e. L_2 cannot exclusively cover any top segment), which implies L_2 is a cover-line loop, a contradiction. This implies that s_{i_1} is a top segment. A similar argument (for L_2, L_3) implies s_{i_2} is a top segment.

We show that $x(s_{i_1}) \leq x(s_{i_2})$. Similar to before, define the subpath P_r^1 of L_1 that goes from e_1 to r_1 and P_r^2 from e_2 to r_2. Considering the two areas of strip S_τ separated by $P_r^1 \cup s_{i_1}$, if segment s_{i_2} is on one side and the entry points o_1, e_2 on the other side, then path P_r^2 must either intersect P_r^1 or s_{i_1}, which is not possible (due to Lemma 7). So, s_{i_2} and e_2, o_2 are on the same part of S_τ cut by $P_r^1 \cup s_{i_1}$. This implies s_{i_2} is to the right of s_{i_1} i.e $x(s_{i_1}) \leq x(s_{i_2})$. □

We can reuse the same arguments in the second part of the proof to conclude the following lemma:

Lemma 31. *Neither of L_1 or L_2 exclusively cover a bottom segment on the right of Γ.*

Proof of Lemma 31. Let L_j be either one of L_1 or L_2. Assume the contrary, that there is some bottom segment s_j to the right of Γ that L_j exclusively covers. So L_3 does not intersect with this segment. Let p_j be the last intersection point of L_j with s_j. Consider the subpath $P_j : p_j \to o_j$ on L_j. Similar to the proof of Lemma 30, we get that both entry points of L_3 are surrounded by $P_j \cup s_j$ from the right or above; which means any top segment that L_3 intersects with, is already intersecting with P_j. This requires L_3 to be a cover-line loop, giving us a contradiction. □

Now we define an alternate path that replaces L_1 and L_2 with two new loops that no longer overlap at Γ, and overall the shadow does not increase but also costs less than the cost of the current solution. The idea of this change (which will be made precise soon) is to follow L_1 from e_1 to the right-most point on L_1 (which must be a reflection on s_{i_1}), then from that point follow a horizontal line until it hits s_{i_2}; if there are portions of L_2 that are above this horizontal line, we follow the upper envlope of those portions of L_2 and the horizontal line (similar to how we reduced the shadow in the case of zig-zag or sink), and then from the intersection point on s_{i_2}, follow the horizontal line back to the right-most reflection on L_1 and continue to follow L_1 to o_1; L_2 is going to be simply replaced with a smaller subset of its projection on the bottom cover-line. We show we will have a cheaper feasible solution with smaller shadow at Γ, a contradiction. Now we describe this more precisely.

Again, let r_1, r_2 be the right-most points on L_1, L_2, respectively. Lemmas 30 and 1 imply that they are reflection points on segments s_{i_1}, s_{i_2}, respectively, which both are top segments. Consider the horizontal line $y = y(r_1)$, and let q be the intersection point of this line with the vertical line $x = x(s_{i_2})$. Define the subpath P_r^2 on L_2 as $P_r^2 : e_2 \to r_2$ (assuming that e_2 is to the left of o_2), In other words, between the two paths from r_2 to the two entry points of L_2, P_r^2 is the one that is above the other. Let U_2 be the portion of P_r^2 in the region bounded by lines $s_{i_1} \cup (y = y(r_1)) \cup s_{i_2}$, and the top cover-line. So these are the portions of P_r^2 that go above the line segment $r_1 q$ (see Figure 25). Let L_1'' be the upper envelope of $U_2 \cup r_1 q$ plus the line $r_1 q$.

So L_1'' consists of a path that goes on the upper envelope of $U_2 \cup r_1 q$ from r_1 to q and then goes straight back to r_1. We now define the replacements for L_1 and L_2.

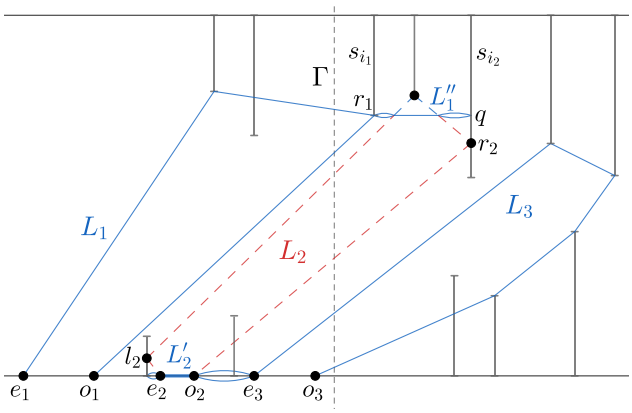


FIGURE 25. Alternative solution for 3 overlapping (non-cover-line) loops. Pairs of arcs represent doubled segments

We replace L_1 with L'_1 as follows:

- Take the subpath $P^1_r : e_1 \to r_1$ on L_1.
- From r_1, follow L''_1 and thus, get back to r_1.
- From r_1, follow the rest of L_1 to o_1.

So L'_1 is obtained by adding L''_1 to L_1 at r_1. If l_2 is the left-most point that L_2 travels, then let L'_2 be a cover-line loop that travels from e_2 left to $x(l_2)$, then right to e_3 and then back to o_2 (this is essentially the projection of the portions of L_2 to the left of e_3 and hence to the left of Γ on the bottom cover-line); recall that we can reduce L'_2 to remove the possible overlapping of its legs. Now replace L_2 with L'_2. We will show that these two loops in total cost strictly less than L_1 and L_2, the shadow does not increase (and in fact shadow decreases at Γ) and we still have a feasible solution. It's clear to see that between the loops L'_1, L'_2, and L_3, only L'_1 and L_3 overlap at Γ. So we decreased the number of overlapping loops at Γ by at least one.

To prove all segments are still covered, note that L'_1 includes the entirety of L_1, and thus covers all the segments that L_1 used to cover. In order to show that all the segments that L_2 covered, are still covered, we only need to show that the segments that L_2 exclusively covered, are still covered. That is because in the new configuration we still have all the parts of L_1 and L_3. According to Lemma 31, there are no bottom segments that L_2 exclusively covers to the right of Γ. Also, it is easy to see that any bottom segment that was exclusively covered by L_2 to the left of Γ must have an x-coordinate between $x(l_2)$ and $x(e_3)$. All of those bottom segments are now covered by L'_2. Finally, for the top segments that L_2 exclusively covers, with the same arguments as in the second part of the proof in Lemma 31, we get that there are no such segments to the left of s_{i_1}. So it suffices to show that only the top segments that L_2 covers to the right of s_{i_1}, are covered. This is easy to see, because L''_1 includes the entire U_2; and it is always on or above L_2 in the range between s_{i_1} and s_{i_2}. Thus, L''_1 will cover all the top segments that L_2 exclusively covers in that range.

Also, the shadow does not increase: the shadow of L''_1 from r_1 to r_2 can be charged to the sections of L_2 between $x = x(r_1)$ and $x = x(r_2)$ and hence is no more than that; note that this portion is entirely to the right of Γ. The cover-line loop L'_2 is entirely to the left of Γ and its shadow can be charged to the shadow of L_2 to the left of Γ in the range $[x(l_2), x(e_3)]$.

Now let's prove that the new cost is decreased compared to L_1 and L_2. L'_1 includes L_1, so we set aside those parts and charge them on L_1. So it suffices to show that L''_1 along with L'_2 can be charged into L_2. Note that L'_2 is part of the projection of L_2 on the bottom cover-line to the left of e_3. So the cost of L'_2 is strictly less than the cost of L_2 to the left of Γ, since L'_2 extends at most to e_3 which is to the left of Γ. As for L''_1, note that L_2 travels back and forth between $x(s_{i_1}), x(s_{i_2})$; so L''_1 can be charged to these two sections of L_2 between $x(s_{i_1})$ and $x(s_{i_2})$.

So at the end, we found a new solution with 1 fewer overlapping loops at Γ, no increase of shadow elsewhere, and with a strictly less cost than OPT_τ. Applying this argument implies that at most two overlapping non-cover-line loops can exist to the left of Γ. So in total on each of the cover-lines of S_τ, there are at most 2 non-cover-line loops to the left of Γ, similarly 2 to the right, plus at most 2 with entry points to opposite sides of Γ. In total, there are at most $6 \times 2 = 12$ overlapping loops at Γ.

Overlapping Ladders. Recall that by Definition 7, ladders are subpaths of OPT_τ (in strip S_τ) that have one entry point on the bottom cover-line of S_τ, and one on the top cover-line. Depending on their orientation compared to Γ, there are two types of ladders (see Figure 26):

- **Type 1 Ladder:** Has both its entry points on the same side of Γ.
- **Type 2 Ladder:** Has its entry points on opposite sides of Γ.

We will prove that there are at most 2 overlapping Type 1 ladders, and at most 5 overlapping type 2 ladders.

Type 1 Ladders. In particular, we show that there is at most one Type 1 ladder with entry points to the left of Γ, and one with entry points to its right. To prove this, assume the contrary, that there are at least 2 overlapping Type 1 ladders with entry points to the same side, say right of Γ. Let L_1 and L_2 be two such ladders.

Let $(b_m, t_m),\ m = 1, 2$ be the entry points of L_m on the bottom cover-line and the top cover-line, respectively. Without loss of generality, assume that t_1 is to the left of t_2. This implies b_1 is also to the left of b_2 (or else L_1 and L_2 intersect inside S_τ). So if we consider cutting S_τ along L_2, L_1 is entirely in one of the two regions created, namely the one that contains b_1, t_1. Since both L_1 and L_2 overlap at Γ and are Type 1, and they both have their entry points on the same side of Γ, say left, this means that both have to

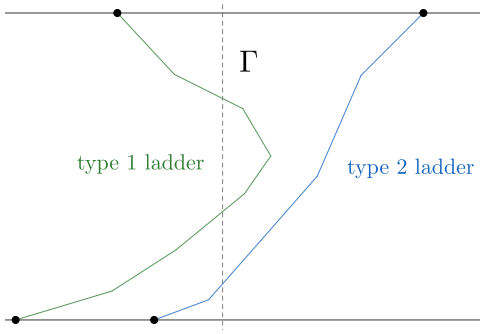


FIGURE 26. An example of type 1 and type 2 ladders

reach to the right of Γ. First, we show that the top-most and bottom-most intersection point of L_1, L_2 with Γ must be on L_2. By way of contradiction, suppose p is a point on L_1 and is the bottom-most intersection of these two ladders on Γ. Consider the subpath of L_1 from b_1 to p, call it L'_1 and consider the region bounded by $L'_1 \cup \Gamma$ and the bottom cover-line, call it A. Since L_2 starts at b_2 inside A and t_2 is outside A, L_2 must either cross Γ at a point lower than p, or cross L'_1, both of which are contradictions. A similar argument shows the top-most intersection point on Γ is with L_2.

Consider any two consecutive crossing of L_1 with Γ, say p_1, p_2, where the subpath of L_1 from p_1 to p_2 (denoted by L'_1) is to the right of Γ. Since L_2 crosses Γ both above and below p_1, p_2 (the lowest and highst intersection points on Γ are with L_2), there is a subpath of L_2 with end-points q_1, q_2 on Γ with q_1 below p_1, p_2, and with q_2 above them, call it L'_2. We consider two cases based on whether L'_2 is on the left or right of Γ, and derive contradictions in each case. If L'_2 is on the right (like L'_1) then L'_1 is inside the region bounded by $L'_2 \cup q_1q_2$ and this violates Lemma 20. So let us assume L'_2 is on the left of Γ. Since p_1, p_2 are between q_1, q_2 there is subpath of L_1 starting from p_1 inside the region $L'_2 \cup q_1q_2$ that crosses q_1q_2. This subpath with L'_2 violates Lemma 20 again. Thus, we conclude that there can be at most 1 Type 1 ladder with entry points to the right of Γ, and similarly, at most 1 with entry points to the left of Γ.

Type 2 Ladders. For each Type 2 ladder L_m with entry points (b_m, t_m) on bottom and top cover-lines, there are two cases:

- b_m is to the left of Γ, therefore t_m is to the right of Γ. We say L_m is a *top-right/bottom-left* ladder.
- b_m is to the right of Γ, therefore t_m is to the left of Γ. We say L_m is a *top-left/bottom-right* ladder.

There can't be two overlapping ladders that one is a top-right/bottom-left ladder, and the other is a top-left/bottom-right ladder (or else they intersect). So if we have a collection of Type 2 overlapping ladders they are all either top-right/bottom-left or all top-left/bottom-right. We show we can have at most 5 Type 2 overlapping ladders. For the sake of contradiction, assume there is a maximal set $\mathcal{L} = \{L_1, L_2, \ldots, L_k\}$ of Type 2 ladders that all overlap at some vertical line Γ with $k \geq 6$ and all are top-right/bottom-left. Let (b_m, t_m), $1 \leq m \leq k$ denote the bottom and top entry points of ladder L_m. Without loss of generality, assume that $x(b_1) \leq x(b_2) \leq \cdots \leq x(b_k)$, which also implies $x(t_1) \leq x(t_2) \leq \cdots \leq x(t_k)$ (or else the ladders will be intersecting each other). Let L^l_m be the subpath of of L_m from b_m to the first intersection of L_m with Γ (so L^l_m is to the left of Γ), and L^r_m be the subpath of L_m from its last intersection with Γ to t_m (so it is to the right of Γ). Note that if $m < m'$ then L^l_m is above $L^l_{m'}$ (in the range that L^l_m is defined) and $L^r_{m'}$ is below L^r_m (in the range that $L^r_{m'}$ is defined) due to Lemma 22. Using Observation 2, this implies L^l_1 covers all the top segments that $L^l_2, L^l_3, \ldots, L^l_k$ cover to the left of Γ and, similarly, L^r_k covers all the bottom segments that $L^r_1, L^r_2, \ldots, L^r_{k-1}$ cover to the right of Γ (we will use this fact shortly).

We will introduce an alternate set of ladders (and loops) that cover all the segments the ladders in $\mathcal{L}$ cover without increasing the shadow anywhere, with a cost strictly smaller cost, and with a smaller shadow at Γ. The set of ladders we introduce differ based on the parity of k. For odd k we keep L_1, L_{k-1}, L_k, and for even k we keep L_1, L_3, L_{k-2}, L_k. We also add some cover-line loops (possibly two copies) to make sure we still have a tour that visits all the points b_j, t_j and all the top and bottom segments that $L_1, \ldots, L_k$ covered remain covered in the new solution [1].

[1]This change is somewhat similar to the proof of patching lemma used in the PTAS for Euclidean TSP that reduces the number of crossings into a region.

Imagine a graph $G(V, E)$ where V consists of all b_j, t_j's and there are edges between two vertices if there is a subpath in OPT between them without visiting any vertex (so we have direct edge between b_j, t_j and also an edge between b_j, t_i if there is a path in OPT between them outside the strip S_τ). Note that G is simply a cycle. In the new alternative solution, we keep L_1, L_k and either L_{k-1} or both L_3, L_{k-2} (depending on the parity of k) and add cover-line loops between some consecutive b_j's and consecutive t_j's such that the resulting graph G' defined based on these new paths still forms an Eulerian (connected) graph on V, all the segments covered in S_τ by $L_1, \ldots, L_k$ are covered. Let b'_2 be the projection of the left-most point on L_2 on the bottom cover-line, and let t'_{k-1} be the projection of the right-most point on L_{k-1} on the top cover-line. We add doubled segment $b_2b'_2$ and $t_{k-1}t'_{k-1}$. These intend to cover any bottom segment exclusively covered by L_2 to the left of b_2, and any top segment exclusively covered by L_{k-1} to the right of t_{k-1}. The doubled segments $b_2b'_2$ and $t_{k-1}t'_{k-1}$ fully appear in the projection of L_2 and L_{k-1} on those cover-lines; meaning that they can be charged onto L_2 and L_{k-1} that travel left (and right) to those segments, respectively. Add each of these two segments twice to the solution. Since we're adding these segments twice, the parity of the degree of nodes in G' won't change. We keep L_1, L_k from $\mathcal{L}$ and add the following segments and ladders as well to the alternative solution (see Figure 27):

- If $k = 2m$ for some integer $m \geq 3$, then include L_3 and L_{k-2}. We also add the following cover-line loops:
  - $b_2b_3,\ b_{k-2}b_{k-1},\ b_{2q-1}b_{2q}$ $(2 \leq q \leq m-1)$
  - $t_2t_3,\ t_{k-2}t_{k-1},\ t_{2q-1}t_{2q}$ $(2 \leq q \leq m-1)$

  We add double the following cover-line loops (i.e. a path back and forth on the same pair of points):
  - $b_{k-1}b_k, b_{2q}b_{2q+1}$ $(2 \leq q \leq m-2)$
  - $t_1t_2,\ t_{2q}t_{2q+1}$ $(2 \leq q \leq m-2)$
- If $k = 2m+1$ for some integer $m \geq 3$, then we include L_{k-2} and also the following cover-line loops:
  - $b_{k-2}b_{k-1},\ b_{2q}b_{2q+1}$ $(1 \leq q \leq m-2)$
  - $t_{2q}t_{2q+1}$ $(1 \leq q \leq m-2)$

  We add double the following segments:
  - $b_{k-1}b_k,\ b_{2q-1}b_{2q}$ $(2 \leq q \leq m-1)$
  - $t_{2q-1}t_{2q}$ $(1 \leq q \leq m)$

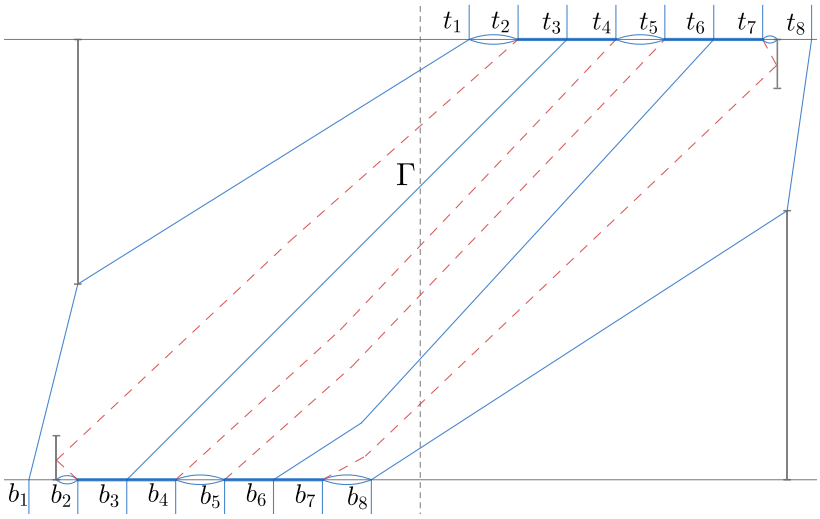


FIGURE 27. Alternative solution for 8 overlapping (bottom-left/top-right) ladders. Red dashed lines are discarded. The arcs represent the doubled segments

It can be seen that with the above additions, if we build the graph G' based on the new paths it is an Eulerian graph as each b_j, t_j has even degree; also G' remains connected since all the $t_1, \ldots, t_{k-1}$ are connected via cover-line loops added at the top and $b_2, \ldots, b_k$ are connected via cover-line loops at the bottom and we have L_1, L_k and there is a path from b_1 to at least one of $b_2, \ldots, b_k$ in outside the strip, and similarly a path from t_k to one of $t_1, \ldots, t_{k-1}$. Thus in the new solution we visit all the points b_j, t_j and this

tour can be short-cut over repeated points to obtain a new solution that visits all the b_j, t_j's and covers all the segments outside the strip S_τ.

Next we show all the segments that $L_1, \ldots, L_k$ were covering, remain covered. Recall that the portion of L_1 to the left of Γ covers all the top segments that were covered by these paths to the left of Γ, and similarly L_k covers all the bottom segments that were covered to the right of Γ. The bottom segments covered to the left of Γ are covered by the new cover-line loops added and similarly, the top segments covered to the right of Γ are covered by the cover-line loops added. So the new solution remains feasible.

Now we are going to bound the total cost of the new solution. We charge all the new parts that we added to some portion of the ladders that we have discarded. Note that in every case, L_2, L_4, L_{k-3}, and L_{k-1} are discarded. We will use only these ladders to charge the new parts to. The doubled segments b_2b_2' and $t_{k-1}t_{k-1}'$ are already charged to the portion of L_2 travelling in the interval $[x(b_2'), x(b_2)]$ and the portion of L_{k-1} travelling in $[x(t_{k-1}), x(t_{k-1}')]$. Now consider the ranges $\beta_j = [x(b_{j-1}), x(b_j)]$, $3 \leq j \leq k$ and $\theta_j = [x(t_{j-1}), x(t_j)]$, $2 \leq j \leq k-1$. These are disjoint, and all β_j's lie under L_2^l and L_4^l, while all θ_j's lie above L_{k-3}^r and L_{k-1}^r. Each of the new included segments (doubled or not) can be charged to one or two of the ladders $L_2, L_4, L_{k-3}, L_{k-1}$.

It can be seen that in the new configuration, there are at most 4 overlapping ladders and 2 overlapping loops (doubled segment loops that we added). This concludes the case for ladders.

In general, when given a collection of loops and ladders, we first alter the ladders as described above (and might get some new cover-line loops in the process), then we apply the alteration on the loops. The statement of Lemma 11 follows from this. □

4.3. **Proof of Theorem 3.** This section is dedicated to the proof of Theorem 3. Let $C_1, \ldots, C_\sigma$ be the cover-lines (as defined in Subsection 2.1) for an instance of the problem with $H \leq 3$. It can be seen that $\sigma \leq 2$; in other words, all the segments of the instance can be covered with only at most 2 cover-lines. If $H \leq 2$, then the number of cover-lines is 1, and it can be seen that the portion on that cover-line itself (doubled from the left-most segment to the right-most segment) is an optimum solution. So let's assume $2 < H \leq 3$, therefore $\sigma = 2$, and that we have a single strip, S_1. Furthermore, there must be both top segments and bottom segments in S_1 (otherwise one of the cover-lines would intersect with all segments). We will essentially prove that the optimum solution must be a bitonic tour.

Take any optimum solution OPT for this instance of the problem, and let p^l and p^r be the left-most and right-most points on it, respectively. There is a path P_1 from p^l to p^r, and there is a path P_2 in the other way. Since OPT is not self-intersecting, and since both P_1 and P_2 cover the range $I = [x(p^l), x(p^r)]$, then for any vertical line Γ with $x(\Gamma) \in I$, they both will intersect with it at distinct points. We can use Lemma 20 (for the concatenation of P_1 and P_2 restricted to the left of Γ) to get that p^l is a right reflection. Similarly, p^r is a left reflection.

Without loss of generality, assume that P_1 includes the upper leg of p^l, and thus P_2 includes its lower leg. Using Lemma 5 for the reflection point p^l and the vertical line $x = x(p^r)$, we get that P_1 is above P_2 in range I, which is the entirety of OPT. Observation 2 implies that all the top segments are covered by P_1, while all the bottom segments are covered by P_2. We claim that there are no reflection points other than p^r and p^l in OPT. To see why this is the case, assume the contrary, that there is some reflection point r on OPT other than those two points.

Without loss of generality, assume $r \in P_1$, and assume that r is the first such reflection point on P_1 after p^l. According to Lemma 6, r is a right reflection. Let s be the segment of the instance that r lies on. If s is a bottom segment, then P_2 will be intersecting with it, and we get a violation of Lemma 7. Thus, s is a top segment.

Now, let $\mathcal{P}_1$ be the concatenation of P_1 (restricted to the subpath from p^l to r) along with the entirety of P_2. $\mathcal{P}_1$ is a path that goes from r (a left reflection on a top segment s) and reaches to the right of s. The rest of the path of P_1 (from r to p^r), refer to it as $\mathcal{P}_2$, is another path that goes from r and reaches to its right. Depending on whether the top leg of r belongs to $\mathcal{P}_1$ or $\mathcal{P}_2$, we get a violation of Lemma 30. This contradiction shows that such r cannot exist, and that both P_1 and P_2 are monotone paths with shadow 1, due to Lemma 3. So in total, OPT has a shadow of 2, as was to be shown. □

Note. It can be shown that in these special cases, we can find an exact solution in poly-time. But since we made some assumptions about the x-coordinates of the segments of the instance, we have to undo them to prove this claim. The resulting algorithm will be somewhat detailed for such a limited special case of the

problem, because we have to cover cases such as vertical legs in an optimum solution. So we only settled on showing that an optimum solution has a constant shadow instead, as it's enough for the purposes of our main theorem in this paper.

5. Extensions and Concluding Remarks

Using Theorem 1, we can get a $(2+\varepsilon)$-approximation for the setting where the input line segments are axis-parallel and have bounded aspect ratio. First we consider the unit-length axis-parallel segments:

Theorem 5. *Given n unit length segments $s_1, s_2, \ldots, s_n$ as an instance of TSPN that are each parallel to either the x-axis or the y-axis, there is a $(2+\epsilon)$-approximation with running time in $n^{O(1/\varepsilon^3)}$.*

Proof. Split the segments into two groups based on them being horizontal or vertical. Let the minimal bounding box for the vertical segments have sides $L_v \times H_v$, and the one for horizontal segments have sides $L_h \times H_h$. Similar to what was mentioned at the start of Section 3, if opt is the cost of an optimum solution OPT, then $\text{opt}/2 \geq \max\{L_v, H_v - 2, L_h - 2, H_h\}$.

Consider the boxes of sizes $L_v \times (H_v - 2)$ and $(L_h - 2) \times H_h$ contained in the aforementioned minimal bounding boxes. Let B be the smallest bounding box that contains these two new boxes. So we get that opt is at least as large as any sides of B; and also it is the case that OPT lies completely inside of B.

The left side of B, refer to it as B_l, either has a vertical segment on it (in the case when the left side of the $L_v \times (H_v - 2)$ box overlaps with B_l), or it has the right-most point of the left-most horizontal line (in the case when the left side of the $(L_h - 2) \times H_h$ box overlaps with B_l). The same argument holds for B_r, the right side of B. If neither of B_l and B_r are on a side of the $(L_h - 2) \times H_h$ box, then it means that all the horizontal segments of the problem have an intersection with the interior of B. In this case, take any horizontal segment s_h that has a length of l_h inside of B; we get that $\text{opt} \geq l_h$.

Assuming $\text{opt} \geq l_h$, take the portion of s_h lying inside B, and break it into $8/\varepsilon$ parts of size $l_h \cdot \varepsilon/8$. For each of these parts, consider their left-most points, and let them be $p_1, p_2, \ldots, p_{8/\varepsilon}$. The OPT must intersect this segment at one of these parts. Assume that p_i is the left-most point of the part that OPT intersects with (we can check all the $8/\varepsilon$ cases). Add p_i to the set of vertical segments, and apply the result of Theorem 1 for parameter $\varepsilon/4$ to get a solution covering all the vertical segments along with point p_i. This solution is a lower bound for the restriction of OPT on s_h along with the vertical segments; with the exception that the intersection on s_h itself can add at most $l_h \cdot \varepsilon/4$ to the cost. So in total, this new solution, along with a doubled copy of the part containing p_i, cost at most $(1+\varepsilon/4) \cdot \text{opt} + l_h \cdot \varepsilon/4 \leq (1+\varepsilon/2) \cdot \text{opt}$. Do the same thing for horizontal segments, meaning find a solution covering p_i and all the other horizontal segments with parameter $\varepsilon/4$ in Theorem 1. We get another solution with cost at most $(1+\varepsilon/2) \cdot \text{opt}$. The conjunction of these two solutions, make a feasible solution for the main problem and cost at most $(2+\varepsilon) \cdot \text{opt}$, proving our claim.

If we don't have $\text{opt} \geq l_h$, it must be the case that there is a point on a horizontal segment on one of the vertical sides of B. This implies that OPT must specifically contain that point. Similar to above, we can add that point to the set of vertical segments and the set of horizontal segments separately; we then find solutions using Theorem 1 with parameter $\varepsilon/2$. Combining those two solutions will yield the same result. □

It is straightforward to extend the result of this theorem to a $(2+\varepsilon)$-approximation for when we have axis parallel line segments of size in $[1, \lambda]$, that runs in time $O(n^{\lambda/\varepsilon^3})$ just as in proof of Theorem 1.

There are several interesting open questions left. First, is Insight 2 actually correct or not? In other words, are there instances for which the shadow of an optimum solution when restricted to a bounded strip is *not* bounded? We have not been able to construct an explicit example extending the picture of Figure 1. Another question is whether one can get a PTAS for the case of axis-parallel line segments or, more generally, when the line segments have only a bounded number of possible directions.

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